COUNSELINK: A WEB-BASED GUIDANCE COUNSELING SCHEDULING AND RECORDS SYSTEM FOR THE DIVISION OF STUDENT AFFAIRS OF MINDANAO STATE UNIVERSITY–MARAWI CITY.

COLLEGE OF INFORMATION AND COMPUTING STUDIES
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By:
GAMPONG, Abdulmalik M.
MACATANONG, Abdensa C.

KHALID, Suhaina A.
Adviser:

MACATOTONG, Amer Hussien T.
Co-Adviser

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CHAPTER 1

1.1 Introduction

Guidance and counseling services are vital in higher education as they promote student well-being, emotional balance, and academic success. According to Corey (2016), counseling is a professional relationship that helps individuals manage stress, make informed decisions, and achieve personal growth. Likewise, Tuazon and Tacuban (2017) emphasized that an organized guidance and counseling program contributes to better student performance, discipline, and adjustment within academic institutions. In the local context, the Guidance and Counseling Information Support System study (2022) also demonstrated that digitalizing counseling services enhances accessibility and responsiveness, particularly in universities that still rely on manual operations. These definitions and findings highlight that effective counseling is an essential service that fosters holistic student development and institutional support.

The Division of Student Affairs (DSA) of Mindanao State University–Marawi City continues to manage its counseling operations manually through paper-based forms and physical logbooks. This leads to misplaced records, mixed-up student files, double-booked appointments, and difficulty tracking counseling sessions and referrals. Counselors also handle students based on assigned colleges, yet matching students to their designated counselor becomes inconsistent, causing delays in sessions. Managing feedback tallies is time-consuming, and referrals requiring psychological expertise are difficult to process due to limited specialist availability. On the students' side, they often experience long waiting times, unknowingly visiting the office when no counselor is available, and have no freedom to choose their preferred counselor. Retrieving counseling results is also inconvenient because they must physically go to the DSA, and communication through phone calls is unreliable. Theoretically, this study is anchored on Vygotsky's Sociocultural Theory, which highlights the role of tools and technology in supporting human activities. Anderson and Dron's concept of digital learning environments further support the use of technology to improve accessibility, efficiency, and communication. Guided by these theories, the study aims to address the existing counseling challenges through a streamlined, technology-driven system.

Despite the importance of counseling in fostering student welfare, the DSA currently struggles with an inefficient and fragmented counseling management system. Manual documentation makes it difficult to maintain accurate records, track student progress, and generate reports for institutional use. Coordination between the DSA and individual colleges is also limited, preventing effective monitoring and follow-up of students in need of guidance. To address these issues, the researchers propose CounselLink: A Web-Based Guidance Counseling Scheduling and Records System that automate appointment scheduling, securely stores counseling records, and provides monitoring access for colleges. By implementing this system, the DSA can improve efficiency, accuracy, and transparency in delivering counseling services, enhancing both institutional productivity and student welfare.

1.2 **Project Context**

This capstone project entitled “CounseLink: A Web-Based Guidance Counseling Scheduling and Records System for the Division of Student Affairs of Mindanao State University–Marawi City” is developed in response to the current challenges faced by the Division of Student Affairs (DSA) in managing guidance and counseling services. At present, the DSA relies heavily on a manual, paper-based workflow for scheduling counseling sessions, recording student interactions, tracking follow-ups, and generating reports. This manual system is prone to errors, including misplaced or lost documents, double-booked appointments, delayed follow-ups, and difficulties in retrieving historical data.

The limitations of the current system not only increase the administrative workload for guidance counselors but also create inefficiencies for students and college representatives. Students often experience long waiting times, difficulty in accessing records, and uncertainty regarding counselor availability. College representatives and administrators face challenges in monitoring student welfare and generating reports for internal tracking and accreditation purposes. These recurring issues highlight the need for a systematic, centralized, and automated solution that aligns with modern technology standards.

The project will be developed within the DSA office of MSU–Marawi City, leveraging the Agile Software Development Methodology. This approach allows the system to be designed, developed, and tested iteratively, incorporating continuous feedback from end-users to ensure the solution meets the actual operational needs of the division. Furthermore, the project is feasible due to the availability of web technologies, support from the university administration, and the readiness of staff and students to adopt digital solutions. By addressing these challenges, the project aims to enhance efficiency, accuracy, and transparency in the university’s guidance and counseling services.

1.3 Purpose and Description

This capstone project entitled “CounseLink: A Web-Based Guidance Counseling Scheduling and Records System for the Division of Student Affairs of Mindanao State University–Marawi City” is to modernize and streamline the guidance counseling operations of the Division of Student Affairs at MSU–Marawi City. The system provides a web-based platform that automates key processes, including appointment scheduling, record management, report generation, and data monitoring, while ensuring security, accessibility, and ease of use for all stakeholders.

The project is designed to address specific operational challenges such as scheduling conflicts, the system allows students to book counseling sessions online, automatically preventing double-bookings and ensuring counselors’ availability is accurately reflected, record accuracy, counselors can maintain digital records of student interactions, ensuring data integrity and facilitating easy retrieval of historical information, administrative efficiency, by automating reports and documentation, the system reduces the time and effort required for manual record-keeping, case tracking, and coordination with college representatives, student accessibility, students can view their counseling history, track appointment status, and receive timely notifications, improving convenience and engagement, accreditation support, CounseLink generates organized reports, consolidated statistics, and digital documentation required for accreditation and quality assurance assessments, aligning with standards set by bodies such as AACCUP and ISO-based evaluations.

The system also strengthens collaboration among stakeholders. College representatives can monitor counseling activities for their students, facilitating better coordination with counselors and supporting student welfare initiatives. The name "CounseLink" reflects the system's core purpose: "Counsel" signifies guidance and support, while "Link" represents the connection between students, counselors, administrators, and colleges through a centralized digital platform.

In conclusion, CounseLink is more than a software system. It is an integrated solution that transforms the counseling workflow, improves operational efficiency, ensures data accuracy and confidentiality, and supports institutional accreditation efforts. By providing a centralized, secure, and user-friendly platform, the system empowers counselors, students, and administrators to achieve a more organized, transparent, and effective guidance and counseling process.

1.4 Objectives of the Study

This capstone project, entitled "CounseLink: A Web-Based Guidance Counseling Scheduling and Records System for the Division of Student Affairs of Mindanao State University–Marawi City," aims to design, develop, and implement a centralized web-based system that automates the scheduling, documentation, and monitoring of counseling services within the Division of Student Affairs.

Specifically, this capstone project aims to:

1. Study the system requirements and user needs through interviews, surveys, and observation of the existing guidance counseling process.
2. Design the database, and interface based on the functional requirements and user feedback.
3. Develop the web-based system with modules for scheduling appointments and record management.
4. Test the system using Alpha, Beta, System Usability Scale (SUS) tool to ensure usability, accuracy, and reliability.

1.5 Scope and Limitations

The scope of CounseLink: A Web-Based Guidance Counseling Scheduling and Records System encompasses the digitalization of guidance and counseling operations under the Division of Student Affairs (DSA) of Mindanao State University–Marawi City. The system covers functions such as online appointment scheduling for counseling and psychological testing centralize counseling record management for secure storage and easy retrieval, automated report generation, messaging and notification services, announcement posting, college monitoring module and RBAC or role-based access control that support multiple user roles including students, counselors, DSA administrators, and college representatives each with specific privileges.

For students, the system provides online scheduling, appointment tracking, viewing of counselor's profile, messaging, viewing of personal counseling history, receiving notifications, and accessing uploaded psychological test results or counselor-issued documents. For counselors, the system offers tools for managing appointments, documenting counseling sessions, storing psychological test results, responding to college requests, generating reports, and responding to student messages. Administrators oversee account creation and management, system configuration, announcement posting, and report generation. College representatives can request counseling data from the DSA and view approved student reports for monitoring and accreditation purposes and perform referral appointment for student.

The system is limited to processes directly related to guidance and counseling services. It does not manage academic records, financial aid, enrollment, student disciplinary cases outside the counseling component, or functions belonging to other university offices. The system also requires a stable internet connection for full functionality, since all data access and transactions are web-based. Access to confidential records is restricted to authorized personnel primarily counselors and the system administrator. Students cannot modify counseling records, and college representatives can only view counseling data upon counselor approval. The system does not replace face-to-face counseling but serves as a digital support tool to enhance documentation, scheduling, and data management.

1.6 Significant of the Study

This study is significant as it contributes to the digital transformation of student support services within Mindanao State University–Marawi City. By integrating an automated platform for counseling management**,** CounseLink promotes efficiency, transparency, and accessibility in handling student counseling records and appointments.

The proposed system will benefit the following stakeholders:

- **Division of Student Affairs (DSA):** Will experience streamlined operations, improved record management, and faster data retrieval for reporting and follow-up.
- **Guidance Counselors:** Can manage appointments, maintain counseling histories, and ensure confidentiality of student information efficiently.
- **Students:** Gain easy access to schedule counseling sessions and ensure that their records are properly managed and securely stored.
- **Colleges:** Can monitor the counseling records of their respective students, provide better academic and emotional support while enhance documentation for accreditation purposes.
- **Institution:** Reinforces MSU's commitment to student welfare and aligns with the university's 9-Point Agenda on Student Welfare and Peacebuilding Initiatives and Digital Transformation, as well as the UN Sustainable Development Goal 17 (Partnerships for the Goals) by promoting collaboration among departments to improve student well-being.
- **Future Researcher:** This study will serve as a reference and baseline for future researchers who wish to enhance, extend, or integrate similar systems related to guidance counseling, mental health services, student information systems, or institutional digital transformation.

CHAPTER 2

Review of Related Literature and Systems

This chapter provides background information relevant to the study, emphasizing the benefits and importance of guidance and counseling services as essential components of education that promote the holistic development of students. It also highlights the significance of integrating technology into the counseling process to enhance accessibility, efficiency, and data management.

2.1 Review of Related Concepts

2.1.1 Digital transformation in Educational Management

The continuous integration of technology in higher education has given rise to the concept which emphasizes the use of computerized systems to enhance administrative efficiency and service quality. Tuazon and Tacuban (2017) explained that the digitalization of student services improves accuracy, accessibility, and responsiveness in school operations. Similarly, Medina (2019) underscored that automating student records promotes transparency and data security, which are crucial for institutions that manage large volumes of student information. These concepts serve as the foundation for CounseLink, which aims to transform the manual counseling process of the Division of Student Affairs (DSA) at Mindanao State University–Marawi City into an efficient and transparent web-based operation.

2.1.2 Guidance and Counseling Information System

Guidance and Counseling Information System (GCIS)**,** which focuses on the integration of technology into counseling management to improve the delivery of student support services. Studies such as Suryanto et al. (2020) and Balmores et al. (2024) have emphasized that web-based counseling systems enable centralized data storage, faster retrieval of student records, and better communication between counselors and students. These systems reduce the reliance on paper-based documentation and enhance the confidentiality and organization of counseling processes. CounseLink adopts the same concept by automating appointment scheduling, session documentation, and reporting, thereby ensuring that the DSA's counseling operations are both efficient and secure.

2.1.3 Security in Information management

This underpins the development of modern educational information systems. DeLone and McLean (2016) highlighted that information quality, system reliability, and user satisfaction are fundamental indicators of a successful system. These principles are critical in handling sensitive counseling data, where confidentiality and controlled access are required. CounseLink applies these information management principles through its secure role-based access control (RBAC) system, ensuring that only authorized DSA personnel and college representatives can view or manage specific student records.

2.1.4 User-Centered Design and Accessibility in Information Systems

This emphasizes that information systems must be intuitive, inclusive, and easy to navigate for all users. Anderson and Dron (2017) stated that technology serves as a medium for collaboration and engagement, allowing users to interact efficiently within digital environments. In line with this, CounseLink is designed to be accessible to various stakeholders. Students, guidance counselors, and college representatives facilitating smooth communication and promoting user adoption. By adhering to user-centered design, the system ensures that users can easily request counseling, manage sessions, and monitor data without technical complexity.

2.1.5 Web-Based System

A web-based system operates through a web browser and connects users to a centralized database via the internet. Skordas, Papadakis, and Kalogiannakis (2017) stated that such systems enhance accessibility, collaboration, and efficiency by allowing users to process and share information in real time. CounseLink applies this concept by providing an online platform that centralizes counseling operations, enabling users to manage appointments, records, and reports efficiently and securely from any location.

2.2 Review of Related Systems

The growing demand for digital transformation in educational institutions has led to the development of numerous systems that enhance counseling and student services. These systems demonstrate the effectiveness of integrating technology into school guidance offices, providing inspiration for the design and implementation of CounseLink**.** The following studies and systems highlight best practices, features, and limitations that guided this project's development.

2.2.1 Development of an Online Counseling System and Usability Evaluation (2011)

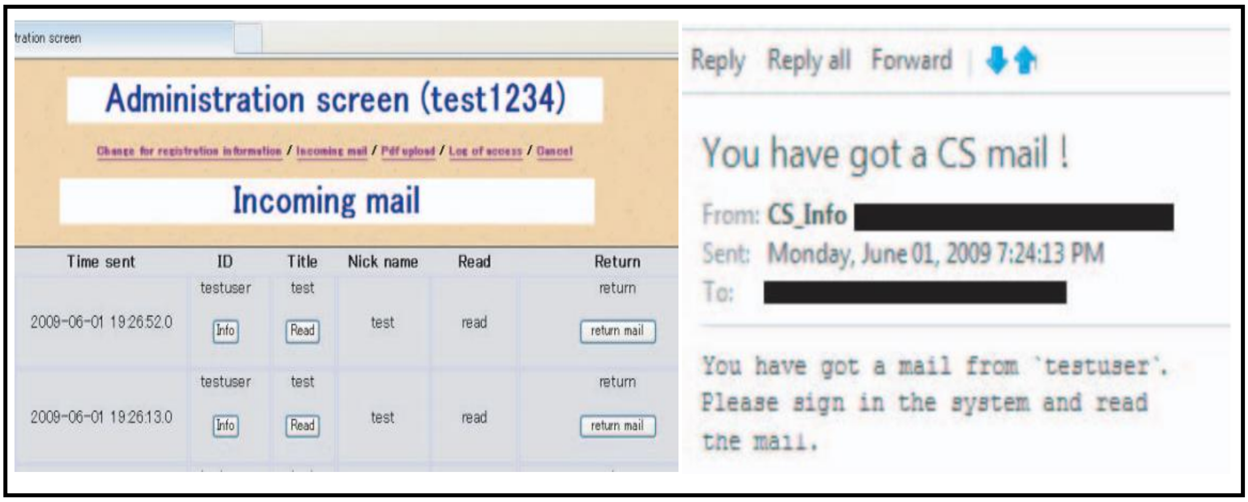

Figure 2.2.1. Online Counseling System

The Online Counseling System (OCS) was an interactive web-based application designed to enable remote communication between students and counselors. It included features such as real-time chat, session recording, and user authentication. The goal was to provide psychological support online, especially for students who were hesitant to seek in-person counseling.

While the OCS improved accessibility and privacy, it was highly communication-oriented and lacked structured documentation of counseling sessions and recordkeeping. CounseLink, on the other hand, focuses on data organization and institutional management, ensuring that every counseling session is documented and stored securely in a centralized database. Instead of real-time chat, CounseLink provides a scheduling system that ensures counselors can efficiently manage multiple appointments while maintaining proper case tracking and documentation

2.2.2 Effectiveness of Guidance and Counseling Services through Web-Based Interactive Media (2024)

Figure 2.2.2. WAHANA BIKONS Web-Based System

This study evaluated the efficiency and user satisfaction of web-based counseling services implemented in selected educational institutions. The findings showed that web-based systems improve communication between students and counselors, increase accessibility to services, and foster a safer and more confidential environment. However, the study also identified challenges such as the absence of integrated data reporting tools and limitations in data sharing between departments.

CounseLink addresses these limitations by integrating an automated report generation feature and allowing college-level monitoring for proper documentation and follow-up. The system ensures that both the Division of Student Affairs and college representatives can coordinate counseling interventions efficiently while maintaining confidentiality.

2.2.3 Guidance and Counseling Record Management System (2021)

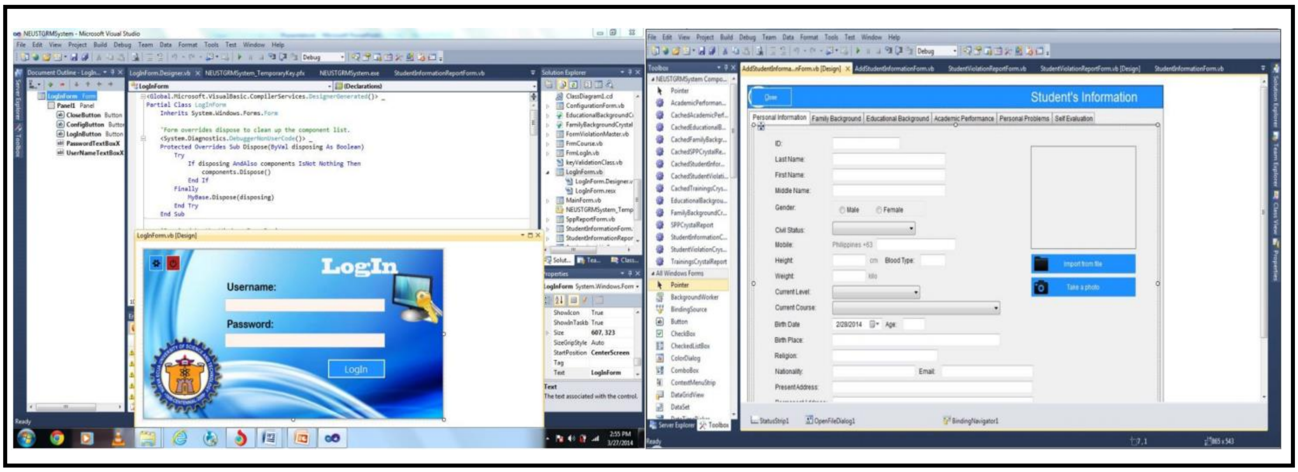

Figure 2.2.3. Guidance and Counseling Record Management System of Nueva Ecija University

The Guidance and Counseling Record Management System, developed for the Nueva Ecija University of Science and Technology, aimed to streamline the management of student records through a software application built using Visual Basic 2017 and Microsoft SQL Server 2017. The system addressed issues in manual record-keeping by enabling users to add, update, and retrieve student information efficiently. It followed the System Development Life Cycle (SDLC) with phases in planning, analysis, design, coding, testing, implementation, and maintenance.

The evaluation of the system demonstrated high ratings in functionality, usability, efficiency, portability, and security**,** confirming its success in automating record management. However, the system was limited to local data storage and lacked features for web-based accessibility and multi-user collaboration**.** CounseLink builds upon this by offering web-based architecture, centralized access, and multi-role functionality (for counselors, students, and college representatives), thus addressing collaboration and scalability needs in a university-wide setup

2.2.4 Web-Based Guidance and Counseling Information System: A Case Study of SMA Negeri 2 Rambah (2024)

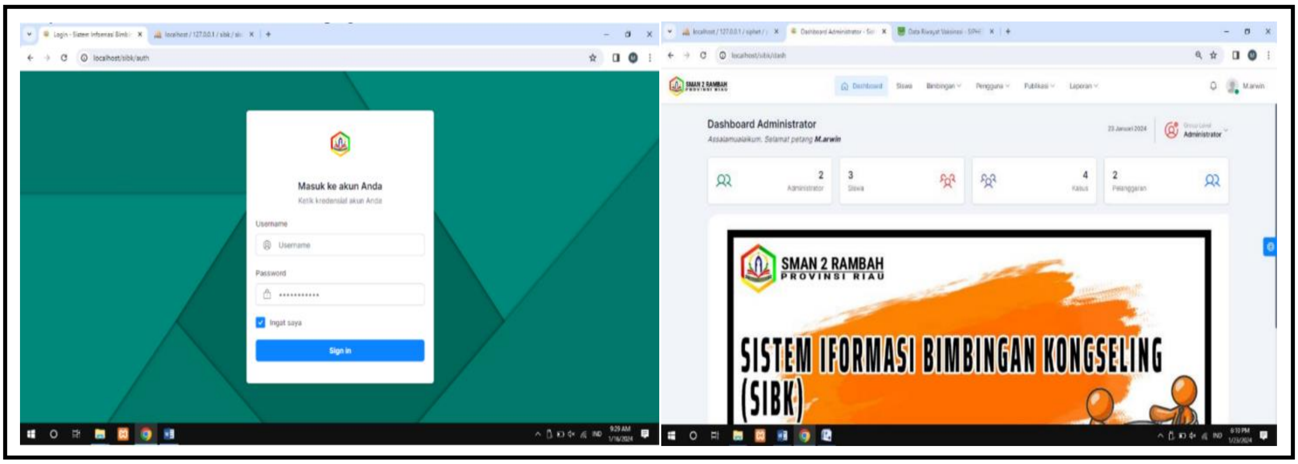

Figure 2.2.4. Sistem Informasi bimbingan Kongseling of SMA Negeri 2 Rambah

The Web-Based Guidance and Counseling Information System developed by Aldo Randra Saputra et al. (2024) for SMA Negeri 2 Rambah was designed to modernize counseling processes and overcome the limitations of paper-based record-keeping. The system was built using PHP and MySQL, integrating modules for student information management, article publication for counseling awareness, and a complaint submission feature that allowed students to report cases of bullying or violence privately.

While this system improved accessibility, security, and communication between students and counselors, it primarily served a high school context with limited reporting and analytics functions. In contrast, CounseLink extends its capability to include **college-**level monitoring, automated reporting, and scheduling management**,** tailored for higher education settings. The proposed system's architecture also integrates role-based access control to ensure that sensitive student information remains secure while still allowing institutional collaboration for academic and psychological support.

2.2.5 Online Individual Inventory and Counseling Information System of CSU–Lasam (2024)

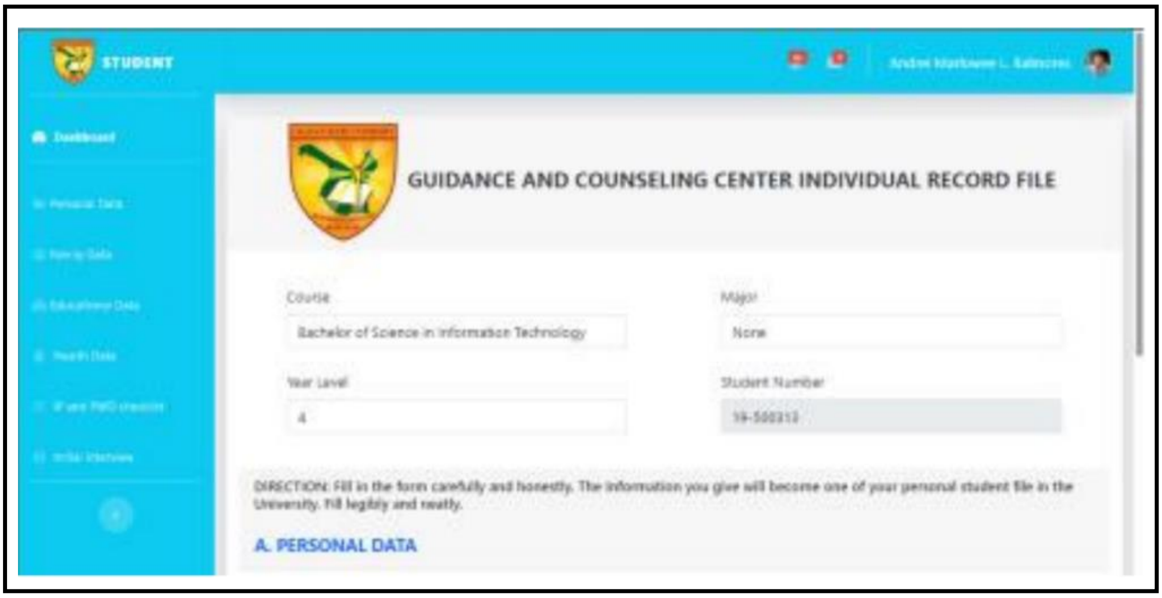

Figure 2.2.5 CSU - Lasam Campus Guidance and Counseling System

This system, developed for Cagayan State University–Lasam Campus**,** was created to automate the management of counseling information and student profiling. It features modules for individual inventory**,** counseling data recording**,** report generation**,** and automated document retrieval**.** The system enables counselors to maintain student profiles digitally and provides an efficient way to track counseling sessions and analyze trends over time.

Although highly functional, the CSU–Lasam system was designed primarily for internal use by counselors, with limited monitoring functions for higher-level departments. CounseLink expands upon this by adding college monitoring and role-based access, ensuring collaboration between the Division of Student Affairs and academic units. This makes CounseLink more adaptive to university-wide implementations, enhancing transparency, coordination, and record security.

2.3 Summary of Related Literature

The literature and systems reviewed collectively emphasize that digital transformation in guidance and counseling services improves accessibility, reliability, and confidentiality of information. Local and national studies confirm the growing importance of web-based solutions in streamlining student services, while international theories validate the integration of technology in educational and psychological support systems.

CounseLink builds upon these findings by incorporating features that address the limitations of manual recordkeeping and fragmented communication within the DSA. By combining technological adoption theories and best practices from previous systems, CounseLink introduces an innovative approach to counseling management that ensures secure data storage, efficient scheduling, and collaborative college monitoring.

Table 2.3 Comparison of Features and Functionalities among Related Systems

Features	Guidance and Counseling Record Management System (2022)	Web-Based Guidance Counseling Information System (2020)	Development of Online Counseling System (2018)	Effectiveness of Web-Based Counseling Services (2019)	Online Individual Inventory & Counseling System of CSU–Lasam (2024)	CounseLink: Web-Based Guidance Counseling Management System Scheduling & Records System
1.Online Appointment Scheduling	✗	✓	✗	✗	✓	✓
2. Counseling Record Management	✓	✓	✓	✓	✓	✓
3.Messeging and notification services	✗	✗	✓	✓	✗	✓
4. Report Generation	✗	✗	✗	✗	✓	✓
5.Announcement Posting	✗	✓	✗	✗	✓	✓
6.Role-Based Access Control (RBAC)	✗	✗	✓	✗	✓	✓
7.College Monitoring Module	✗	✗	✗	✗	✗	✓

2.4 Technical Background

The Division of Student Affairs (DSA) of Mindanao State University–Marawi City is an essential unit that ensures the welfare and development of students. It manages services such as guidance and counseling, student assistance, discipline, and psychosocial support in relation to academics, catering to around 700 students each semester. The Guidance and Counseling section handles individual counseling sessions, psychological testing, and documentation, while coordinating with colleges for student monitoring and accreditation.

At present, counseling operations are done manually. Students seeking counseling through walk-in, referral, or initial appointments, fill out forms and wait for the counselor to schedule sessions. Records, session updates, and follow-ups are handwritten and stored physically. The same manual approach applies to psychological testing and shifting procedures, where students coordinate personally and results are processed on paper. For accreditation, student data are shared through Google Drive, leading to fragmented record management. This manual process causes inefficiencies such as misplaced files, scheduling conflicts, and delays in information sharing. Counselors spend more time on paperwork instead of providing student support, and coordination with colleges is often slow and inconsistent.

To address these challenges, the researchers propose CounseLink: A Web-Based Guidance Counseling Scheduling and Records System for the DSA. The system will automate appointment scheduling, centralize counseling records, and integrate college-level monitoring. Through digitalization, CounseLink aims to enhance efficiency, data accuracy, and accessibility, strengthening the delivery of guidance and counseling services across the university.

2.4.1 Organizational Structure

Figure 2.4.1 illustrate the Organizational Structure of the Division of Student from director to the administrative affairs and guidance section.

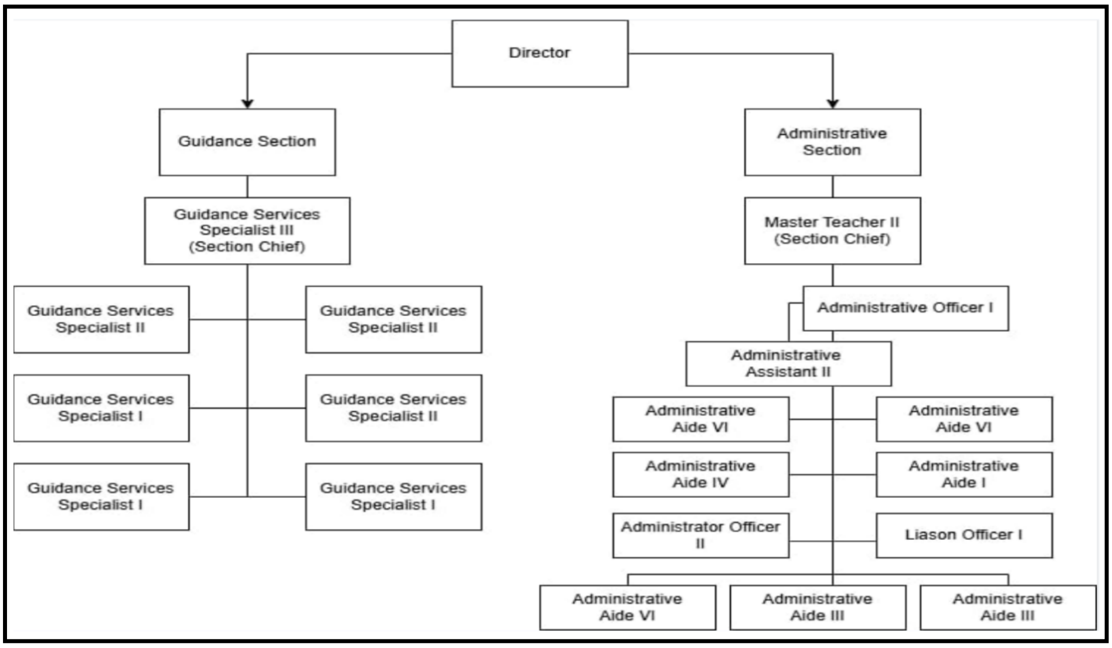

Figure 2.4.1: Organization Chart of the Division of Student Affairs

The Division of Student Affairs (DSA) is led by a Director who oversees two key sections: the Guidance Section and the Administrative Section, each headed by a Section Chief. The Guidance Section, led by the Guidance Services Specialist III, provides counseling and psychological services, supported by Specialists II and I who handle sessions, records, and referrals. The Administrative Section, supervised by Master Teacher II, manages clerical, logistical, and documentation tasks. It includes administrative officers and aids responsible for recordkeeping, communication, and coordination. The Liaison and Administrative Officers also assist in external communication, report preparation, and event coordination for the DSA

2.4.2 Current Workflow

These diagrams below show the current workflow of the services in guidance section such as Counseling and psychological testing start by going at the DSA office and requesting a counseling (walk-in, referral or initial) service or psychological test. As well as how each colleges request counseling data report for college purposes, such as accreditation.

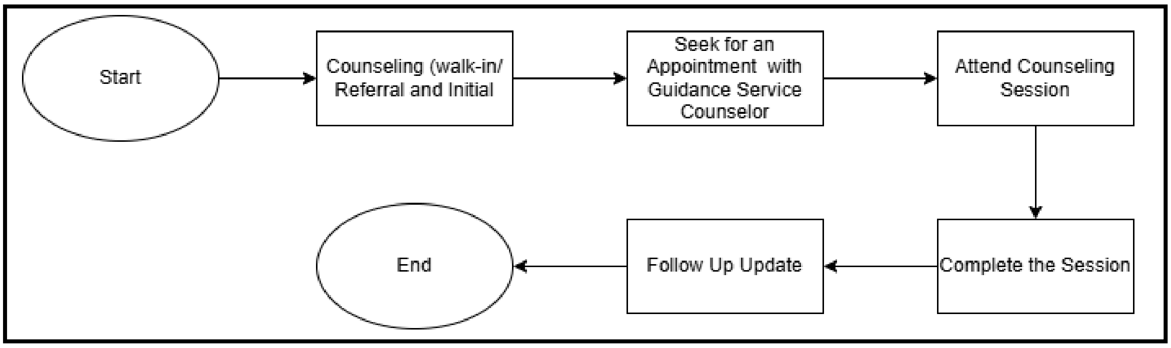

Figure 2.4.2. Workflow of Requesting Counseling

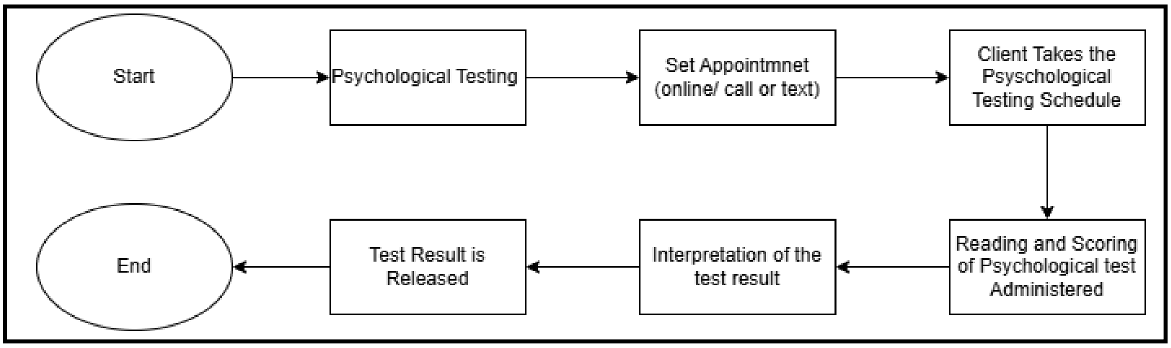

Figure 2.4.2. Workflow of Requesting Psychological Testing

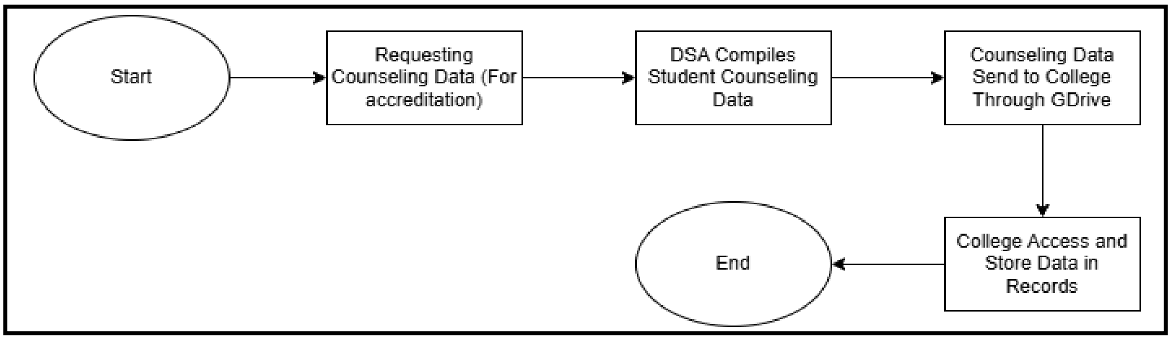

Figure 2.4.2. Workflow Requesting Counseling Data for Accreditation

2.4.3 Proposed Workflow

This Diagram below shows the proposed workflow of how users log in to the system and need to have valid credentials, how students request appointments, request psychological testing how, college representative request counseling report and how counselors confirm pending requests and how each request and transaction are being processed.

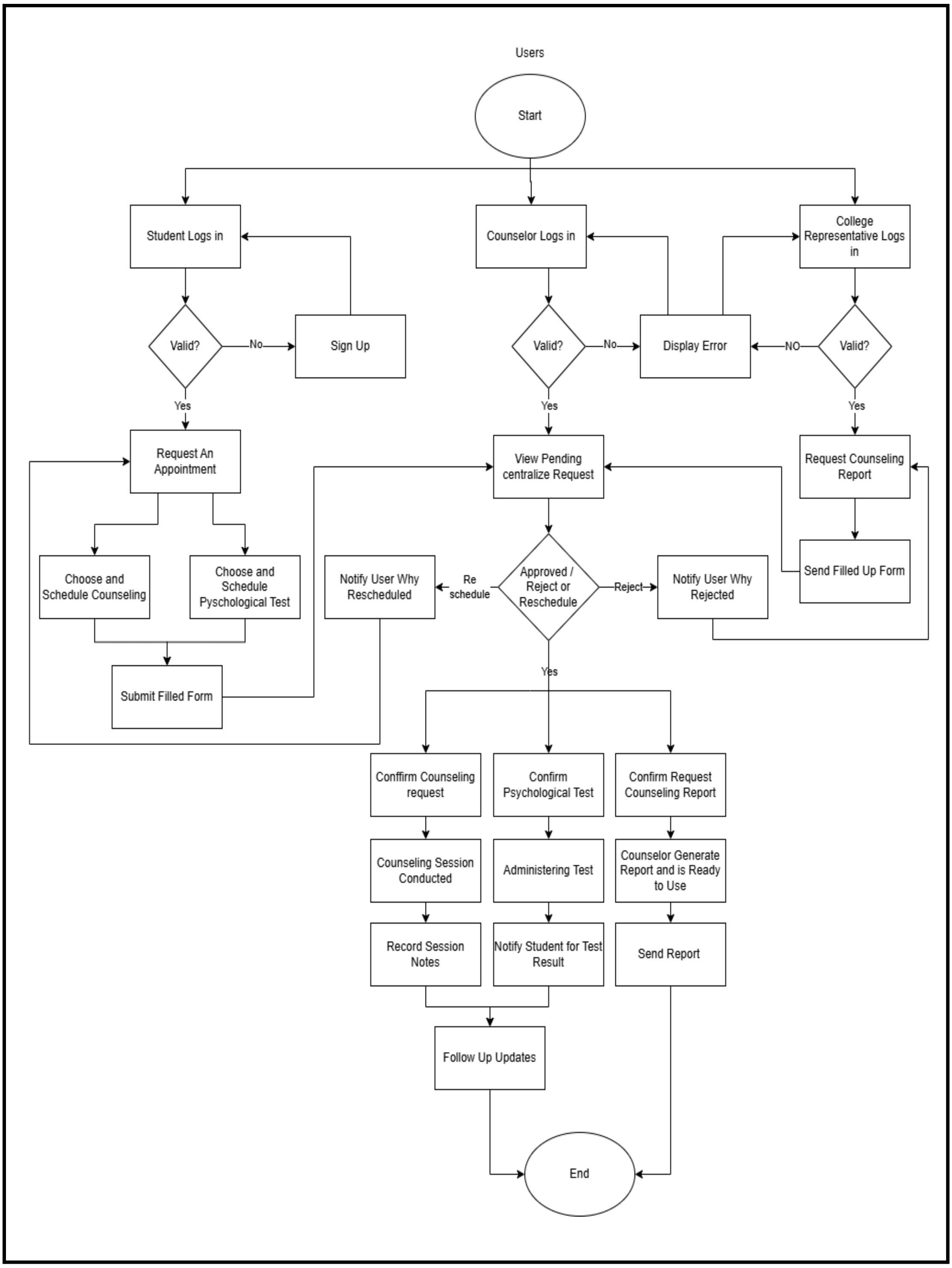

Figure 2.4.3. Proposed Workflow for the Guidance Counseling of DSA

CHAPTER 3

METHODOLOGY

This chapter presents the methods and techniques employed by the researchers in conducting the study and developing the CounseLink: A Web-Based Guidance Counseling Scheduling and Records System for the Division of Student Affairs of Mindanao State University–Marawi City**.** The chapter discusses the development methodology, data gathering techniques, analytical tools, and the overall framework used to ensure that the system aligns with the needs of the Division of Student Affairs (DSA). The Agile "Scrum" Methodology is adopted due to its iterative, flexible, and user-centered approach, which is highly suited for system development requiring continuous feedback and refinement.

Data gathering procedures—such as interviews, observations, and document reviews were utilized to identify the existing problems in the current counseling process of the DSA. The information gathered guided the analysis, design, development, and testing of the system.

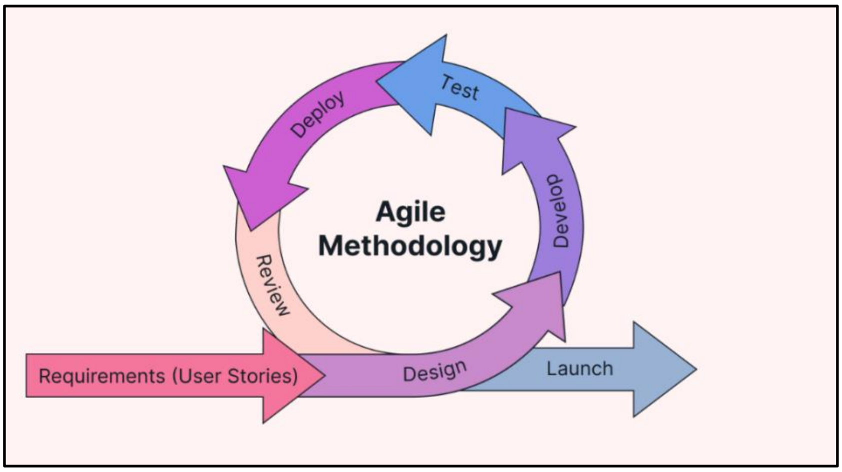

Figure 3: Agile "Scrum" Methodology

3.1 Requirements Analysis

The proposed system, "CounseLink: A Web-Based Guidance Counseling Scheduling and Records System for the Division of Student Affairs of Mindanao State University–Marawi City," aims to automate counseling appointments, record-keeping, psychological test management, and college monitoring. During this phase, the researchers gather user requirements through interviews, surveys, and observation of the existing manual processes to identify the exact needs of counselors, students, and college representatives.

3.1.1 Empathy Map

These figures below present the Empathy Map used to better understand the perspectives, needs, and experiences of the system's primary users, the students, counselors, college representatives, and administrators. The Empathy Map analyzes what users say, think, do, and feel in relation to the existing manual counseling process. By identifying their frustrations, motivations, and expectations, the researchers gain deeper insights that guide the design of a user-centered system aligned with real-world needs and behaviors.

SAYS

- “I need help but I don’t know how to start the counseling process.”
- “It’s hard to know when the counselor is available.”
- “I hope my information stays confidential.”
- “I wish scheduling was easier and faster.”
- “I don’t want to keep going back to the office just to follow up.”
- “I’m afraid my record might get lost or mixed up.”

THINKS

- “Will the counselor understand my situation?”
- “What if my classmates find out I am going to counseling?”
- “I am worried my schedule might conflict with classes.”
- “Is the DSA keeping accurate records about my sessions?”
- “Why does the process take so long?”
- “I hope I get updates without needing to visit the office.”

STUDENTS

DOES

- Visits the DSA office to request counseling.
- Fills out the physical counseling forms.
- Waits for the counselor to manually check availability.
- Returns repeatedly for follow-up announcements or results.
- Communicates through text, call, or walk-in for psychological testing schedules.
- Seeks assistance from college representatives for accreditation documentation.

FEELS

- Anxiety, due to personal problems or long waiting times.
- Frustration, because of manual processes and unclear updates.
- Embarrassment, fear of being seen by peers at the office.
- Relief, once counseling is finally scheduled or completed.
- Confusion about the steps or required documents.
- Worry, about confidentiality of records.

Figure 3.1.1.1 Empathy Map of a Student

SAYS

- “I need to manage student counseling requests efficiently.”
- “I must track all appointments and histories.”
- “I want an organized view of student records.”
- “Student files are too many to manage manually.”
- “It’s hard to schedule sessions efficiently.”
- “I want a get report faster”
- “It’s hard to do tallying.

THINKS

- “Are students getting the help they need?”
- “I need a clear system to avoid scheduling conflicts.”
- “This should help me lessen paperwork.”
- “Communication needs to be fast and clear.”

COUNSELOR

DOES

- Check physical records and folders.
- Schedules sessions manually.
- Write reports and summaries by hand.
- Coordinates with colleges for student concerns.

FEELS

- Pressured if many requests come in
- Responsible for student wellbeing
- Satisfied with an organized digital system
- Stressed if system is slow or confusing
- Concerned about record accuracy.

Figure 3.1.1.2: Empathy Map of The Counselor

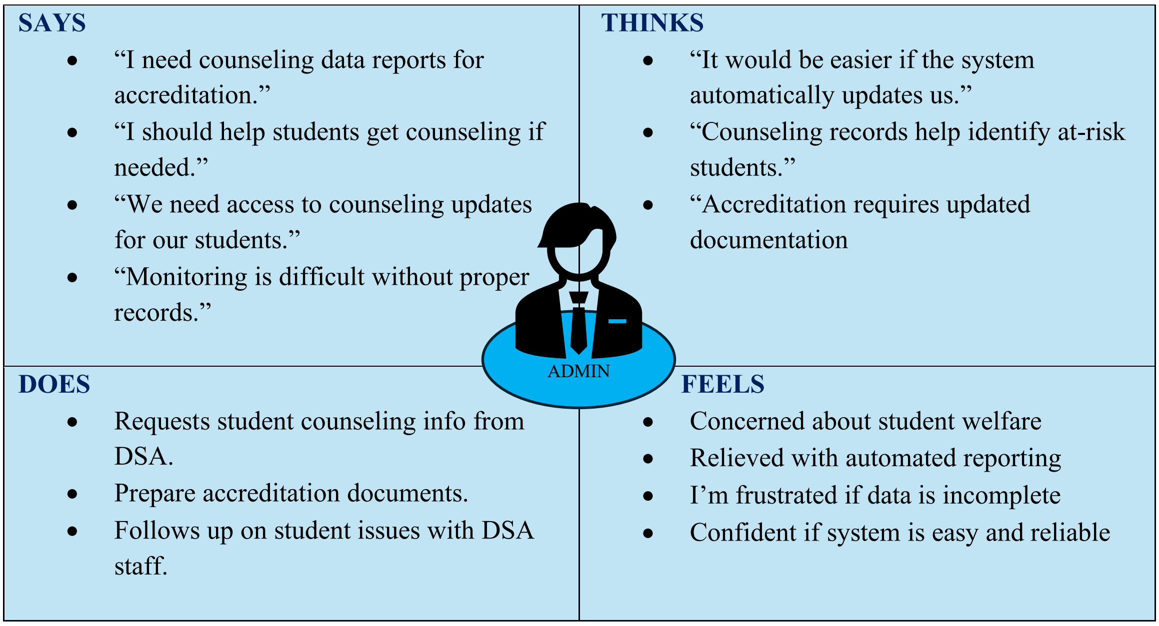

Figure 3.1.1.3: Empathy Map of a College Representative

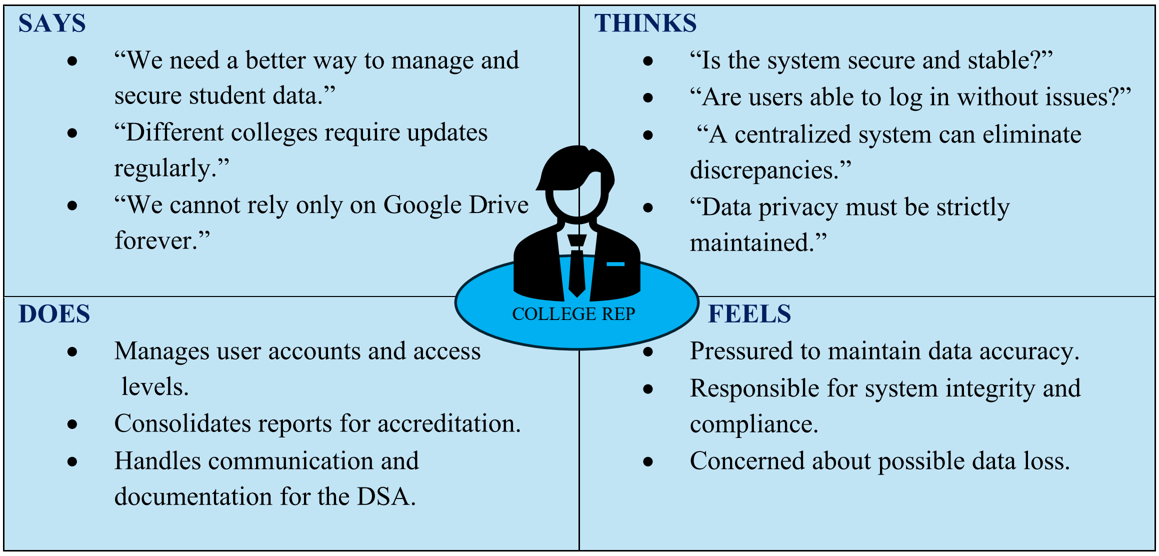

Figure 3.1.1.4: Empathy Map of the Administrator

3.1.2 Functional Requirements and Non-functional Requirement

Table 3.1.2.1 below identifies and defines the essential requirements needed for the development of the CounseLink system. The functional requirements specify the core features, behaviors, and actions that the system must perform to support the workflows of students, counselors, college representatives, and administrators. Meanwhile, the non-functional requirements describe the quality attributes and performance standards the system must meet, including security, usability, reliability, and responsiveness. Together, these requirements ensure that the system operates effectively, securely, and in alignment with the needs of the Division of Student Affairs.

Table 3.1.2 Functional and Non- Functional of the System

Requirement	Type	Priority
The system shall allow users (Student, Counselor, Admin, College Rep) to log in and log out securely	**Functional**	**Mandatory**
The system shall allow students to sign up and create an account	**Functional**	**Mandatory**
The system shall allow all users to view and edit their profile information	**Functional**	**Mandatory**
The system shall allow students to request counseling appointments and submit preferred schedules	**Functional**	**Mandatory**
The system shall allow counselors to review and manage appointment requests (Accept, Reject, Reschedule)	**Functional**	**Mandatory**
The system shall allow students to request psychological tests	**Functional**	**Mandatory**
The system shall allow Students to view and download their psychological test results	**Functional**	**Mandatory**
The system shall allow counselors to add, update, delete, and view student counseling session records	**Functional**	**Mandatory**
The system shall allow counselors to generate counseling reports	**Functional**	**Mandatory**
The system shall allow Counselors to generate counseling reports.	**Functional**	**Mandatory**
The system shall allow Counselors to send counseling reports to College Representatives.	**Functional**	**Mandatory**
The system shall allow College Representatives to request student counseling data and referrals.	**Functional**	**Mandatory**
The system shall allow College Representatives to view, print, and save shared counseling reports.	**Functional**	**Mandatory**
The system shall allow DSA Administrators to create, update, and deactivate Counselor and College Rep accounts.	**Functional**	**Mandatory**
The system shall allow DSA Administrators to post system announcements.	**Functional**	**Mandatory**
The system shall maintain an audit log of critical actions (uploads, edits, deletions, approvals).	**Functional**	**Mandatory**
The system shall send automated notifications for appointment updates and psychological test updates.	**Functional**	**Mandatory**
The system should provide internal messaging between Students and Counselors.	**Functional**	**Optional**
The system shall restrict access using Role-Based Access Control (RBAC).	**Non-Functional**	**Mandatory**
The system shall encrypt all user passwords during storage and transmission.	**Non-Functional**	**Mandatory**
The system dashboard shall load within 3 seconds under normal usage.	**Non-Functional**	**Mandatory**

The system shall support at least 50 simultaneous users without performance degradation.	**Non-Functional**	**Mandatory**
The system interface shall be responsive and adapt to different screen sizes.	**Non-Functional**	**Mandatory**
The system shall perform regular scheduled backups of all databases and user data.	**Non-Functional**	**Mandatory**
The system shall automatically log out users after a period of inactivity (session timeout).	**Non-Functional**	**Mandatory**
The user interface should be simple, clean, and user-friendly for all user types.	**Non-Functional**	**Optional**
The system should be maintainable and modifiable for future improvements.	**Non-Functional**	**Optional**

3.1.3 Gantt Chart

Table 3.1.1 below is the Gantt chart of this system. It illustrates the journey of each phase from the start and finish. As illustrated below, every phase has its corresponding implementation phases. The project began in August, and the proponent divided the months from August to July to finish the capstone project.

Table 3.1.1 CounseLink Management System Gantt Chart

SDLC ACTIVITIES	AUG	SEP	OCT	NOV	DEC	JUN	FEB	MAR	APR	MAY	JUNE	JULY
PLANNING												
Data Gathering and Client Interview												
Define scope and objectives												
Collect initial Data and Analyze it												
Develop project plan												
ANALYSIS												
Gather requirements												
Analyze current system												
DESIGN												
Hierarchical Input-Process-Output (HIPO) model												
Input-Process-Output (IPO) model												
Use case Diagram												
Entity-Relationship Diagram (ERD)												
Developing System Architecture												
CODING												
Setup development environment												
Frontend development												
Backend development												
Integration												
TESTING												
Integration testing												
Alpha Testing												
Beta Testing												
User Acceptance Testing (UAT)												

3.2 DESIGN

This will focus on the data requirement, software construction and the interface of the system. The researchers will follow the requirements specified by the Counselink management system and make use of the different tools such as hierarchical input-process-out, input-process-output diagram, use case diagram, entity-relationship diagram (ERD), and architectural design.

3.2.1. HIERARCHICAL INPUT-PROCESS-OUTPUT

Figure 3.2.1 below section presents the Hierarchical Input–Process–Output (HIPO) model, which provides a structured overview of the major functions of the CounseLink system. The HIPO diagram breaks down the system into its main components and sub-processes, illustrating how inputs from users are transformed into outputs through specific operations. By organizing the system into a clear hierarchy, the HIPO model helps define the overall workflow, clarify responsibilities of each module, and guide the development of the system's architecture. This ensures that all processes from appointment scheduling to report generation are logically connected and aligned with the functional requirements of the Division of Student Affairs.

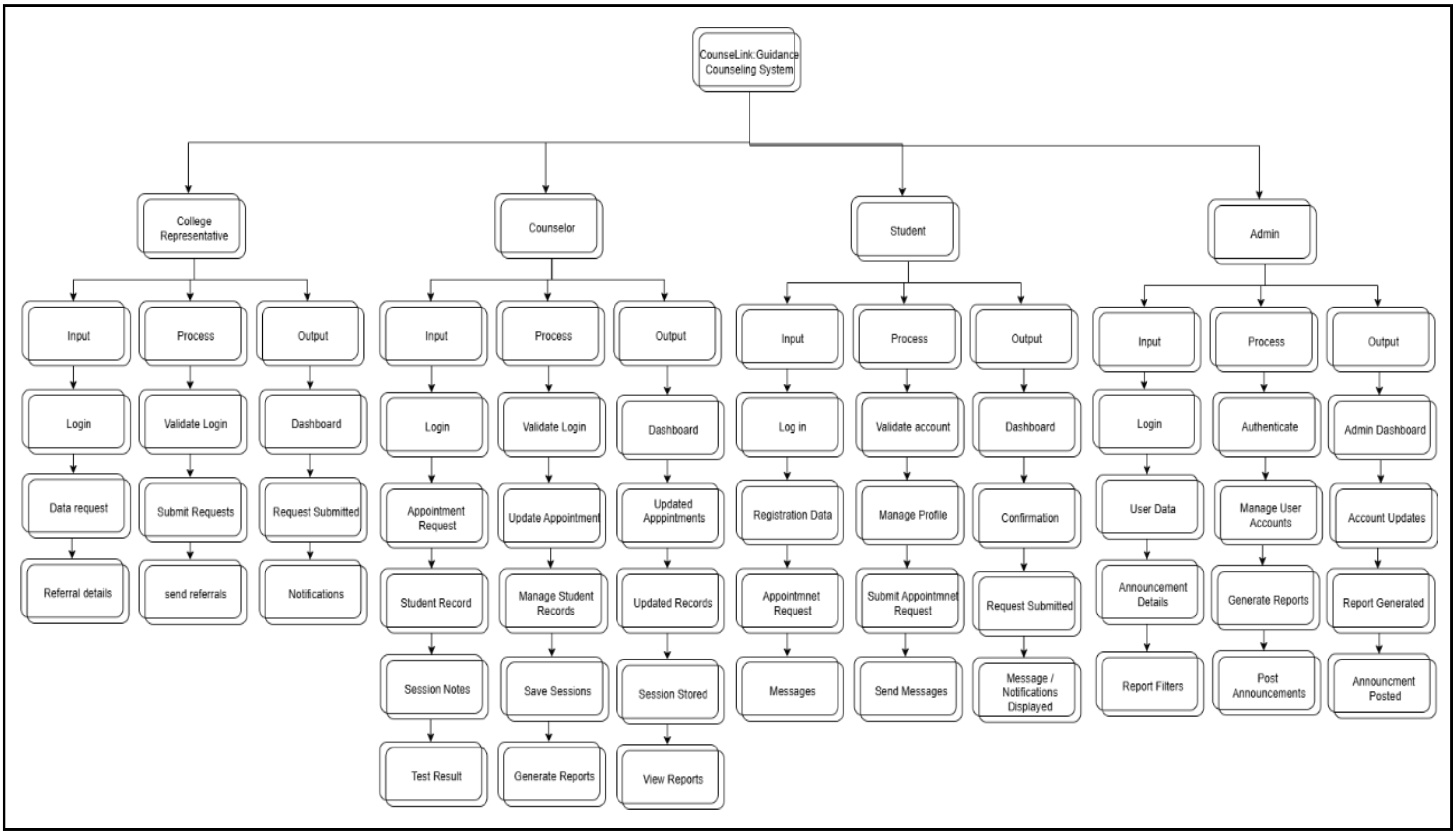

Figure 3.2.1 Hierarchical Input-Process-Output

3.2.2 Input-Process-Output

Below show the categorized Input-Process-output of the proposed system entitled "Counselink: A Web-based Guidance Counseling Scheduling and Records System for the Division of Student Affairs of Mindanao State University–Marawi City**".** The Input-process-output has been categorized in four parts which are admin, counselor, student, & college representative.

3.2.2.1 ADMIN

Figure 3.2.2.1 presents the IPO model for the Admin, focusing on the administrative operations that maintain the overall system environment. The admin role is responsible for managing user accounts, configuring system settings, posting announcements, and monitoring system activity. This IPO model defines the inputs provided by administrators, the internal processes executed by the system, and the resulting outputs. This ensures that administrative functions are clearly structured and aligned with system management requirements.

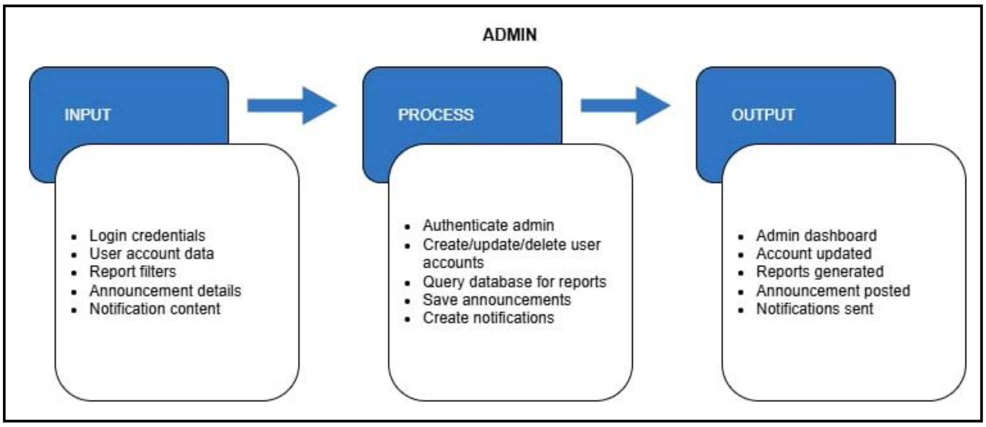

Figure 3.2.2.1 Input-Process-Output (ADMIN)

3.2.2.2 STUDENT

Figure 3.2.2.2 describes the IPO model for the student module, which outlines how students interact with the system to access and utilize counseling services. Students provide inputs such as registration details, appointment requests, message inquiries, and test result viewing actions. The system processes these inputs by validating information, scheduling appointments, retrieving records, and sending notifications. The output includes confirmed appointments, displayed counseling history, accessible psychological test results, and received updates. The Student IPO model ensures that the system remains user-friendly, accessible, and responsive to students' counseling needs.

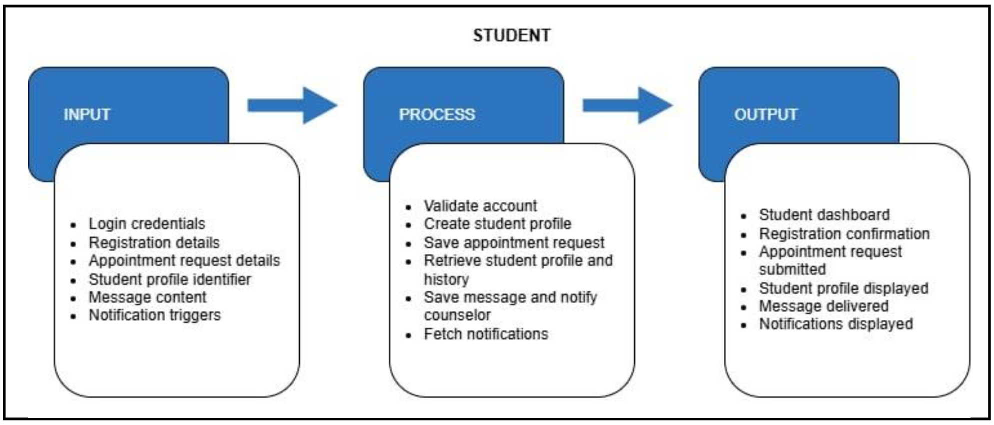

Figure 3.2.2.2 Input-Process-Output (STUDENT)

3.2.2.3 COUNSELOR

Figure 3.2.2.3 details the IPO model for the Counselor module, highlighting the critical role counselors play in managing student cases. Counselors provide inputs such as session notes, appointment decisions, psychological test uploads, and generated reports. The system processes these actions by storing records securely, updating appointment statuses, generating summaries, and routing reports to authorized parties. The output includes documented counseling sessions, processed student requests, released test results, and shared reports. The Counselor IPO model supports efficient and organized counseling management to improve service delivery.

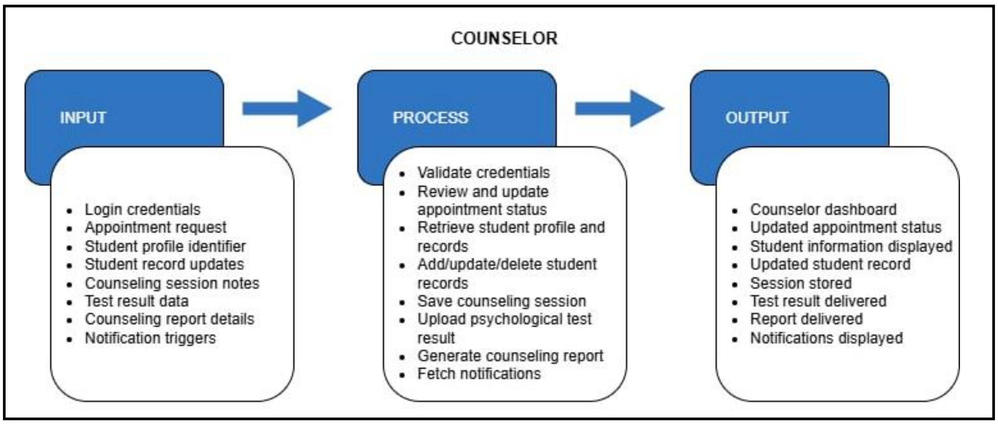

Figure 3.2.2.3 Input-Process-Output (COUNSELOR)

3.2.2.4 COLLEGE REPRESENTATIVE

Figure 3.2.2.4 below outlines the IPO model for the College Representative module, which defines how representatives request and access counseling information for their students. Inputs include data requests, referral submissions, and document retrieval actions. The system processes these inputs by verifying permissions, retrieving appropriate records, and preparing downloadable reports. The output includes approved counseling documents, retrieved reports for accreditation, and updated communication logs. The College Representative IPO model ensures proper coordination between the DSA and academic units while maintaining controlled and secure access to student information.

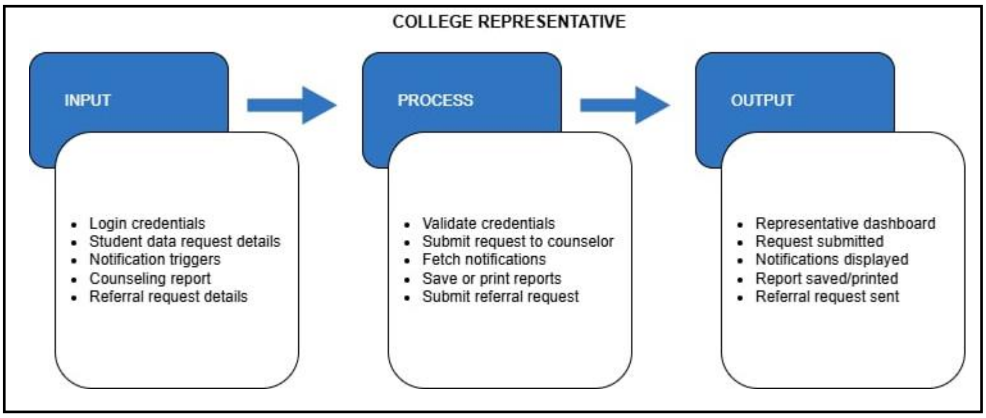

Figure 3.2.2.4 Input-Process-Output (COLLEGE REPRESENTATIVE)

3.2.3 Use Case Diagram

Figures below present the Use Case Diagrams of the CounseLink Management System, for each user role, illustrating the interactions between users and the system. These diagrams show the specific actions each role can perform and help define the system's functional boundaries and user responsibilities.

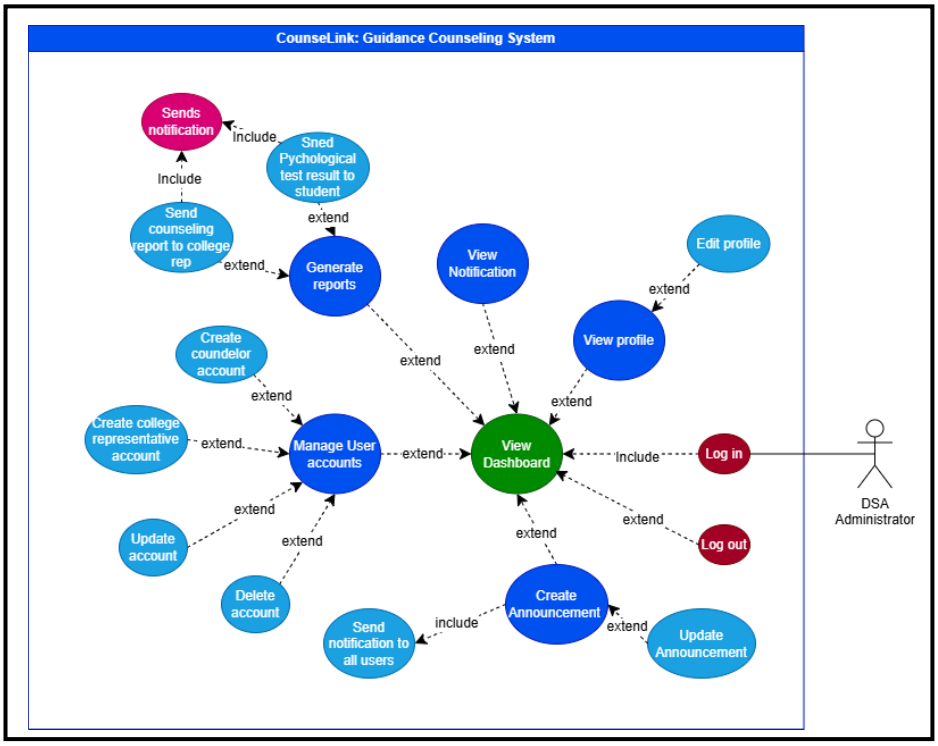

Figure 3.2.3.1 ADMIN Use Case Diagram

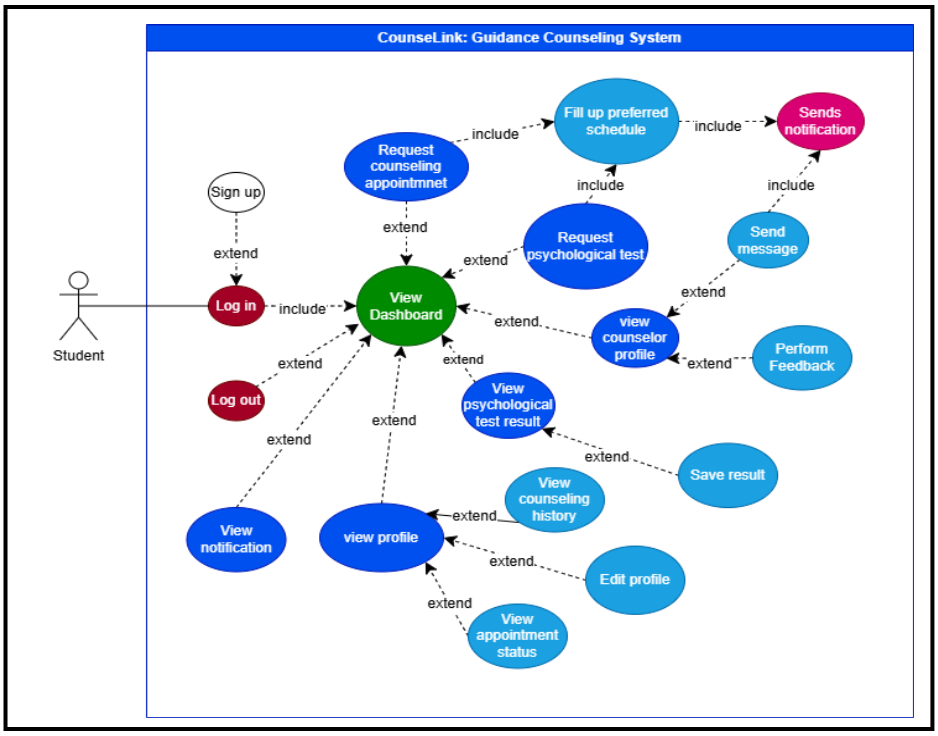

Figure 3.2.3.2 STUDENT Use Case Diagram

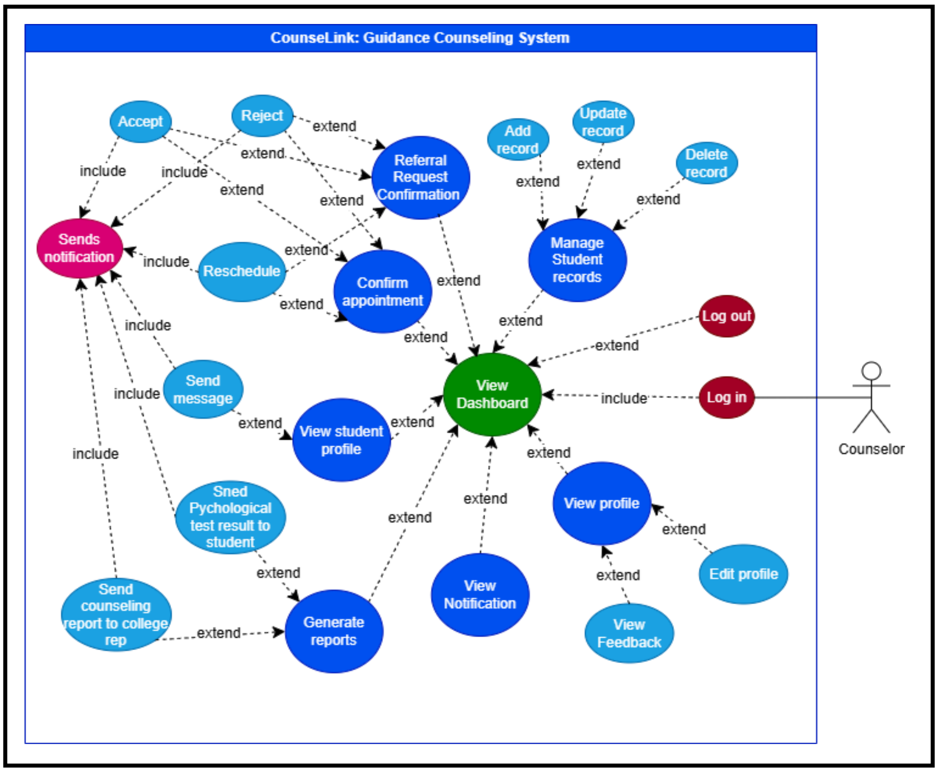

Figure 3.2.3.3 COUNSELOR Use Case Diagram

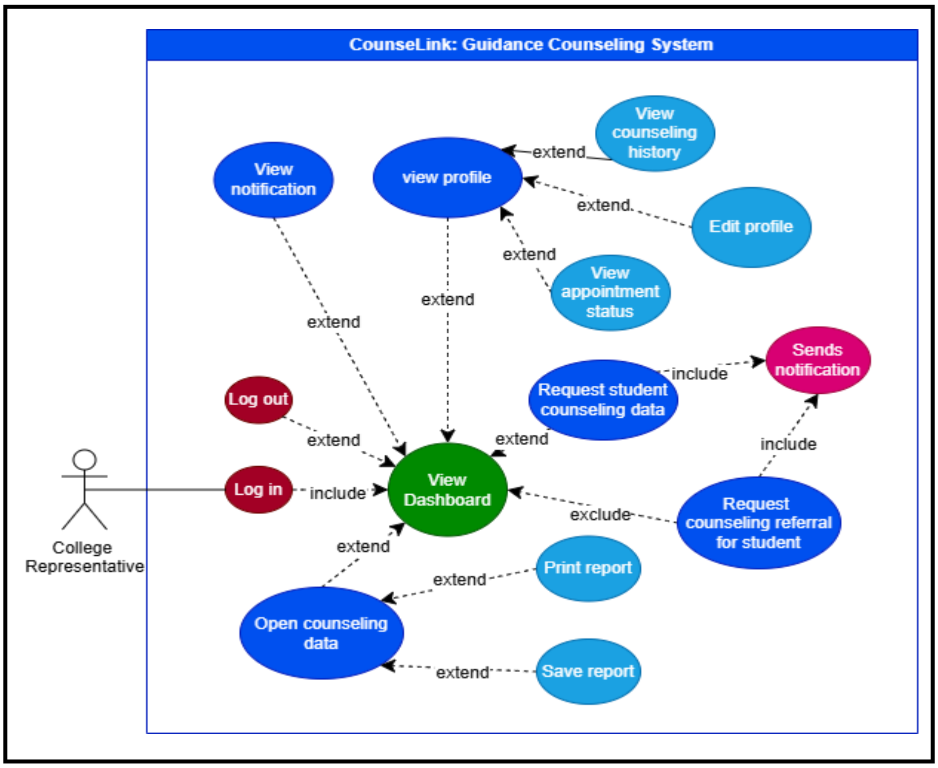

Figure 3.2.3.4 COLLEGE REPRESENTATIVE Use Case Diagram

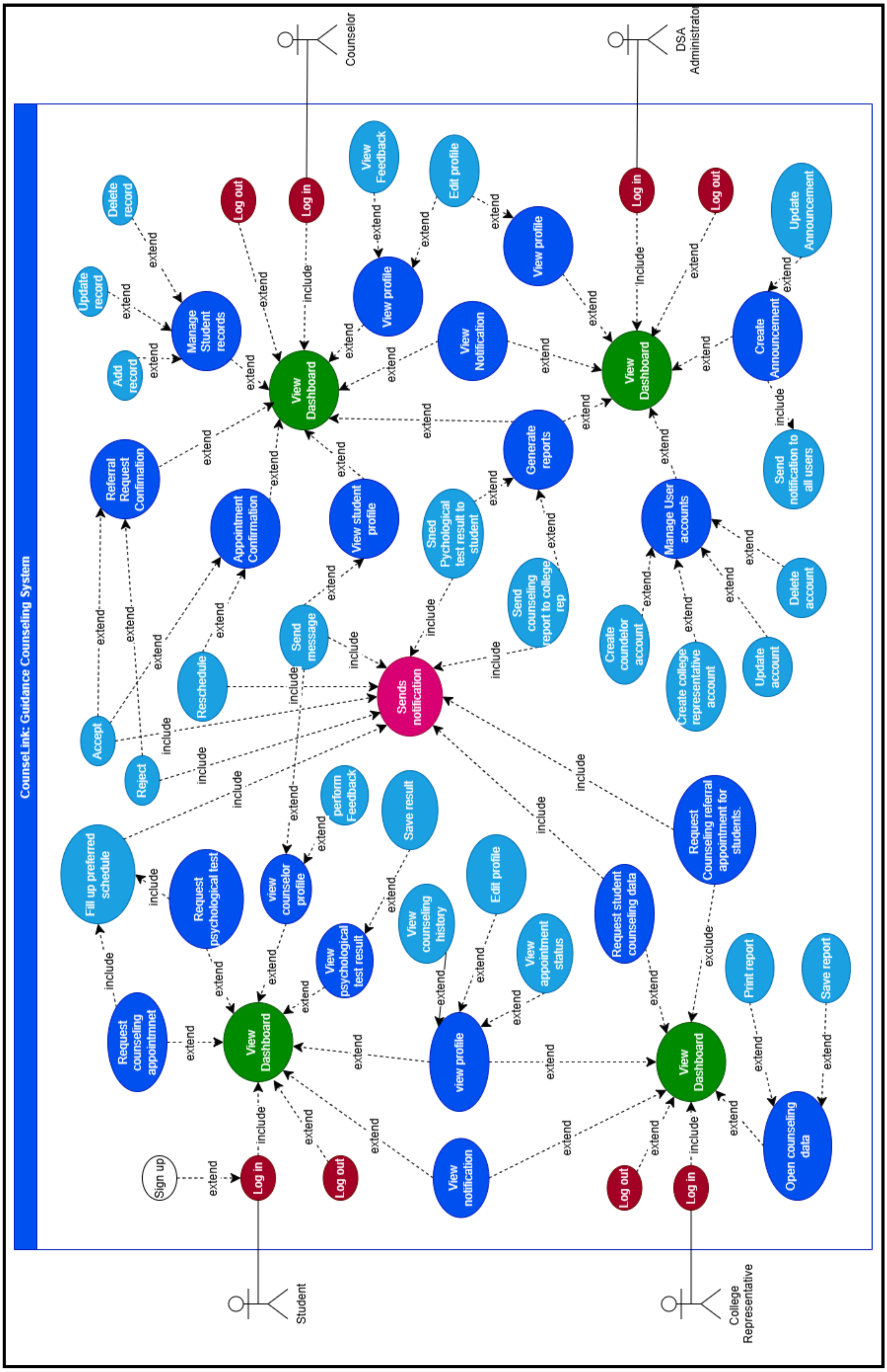

Figure 3.2.3.5: Overall Use Case Diagram

3.2.4 Entity-Relationship Diagram

The Entity–Relationship Diagram (ERD) of CounseLink presents the logical structure of the database by identifying the main entities, their attributes, and the relationships that connect them. It shows how users, appointments, counseling records, psychological test results, report, psychological test result, announcement and message are linked in a centralized schema, ensuring that all guidance and counseling data are stored consistently and can be retrieved efficiently for scheduling, monitoring, and reporting purposes.

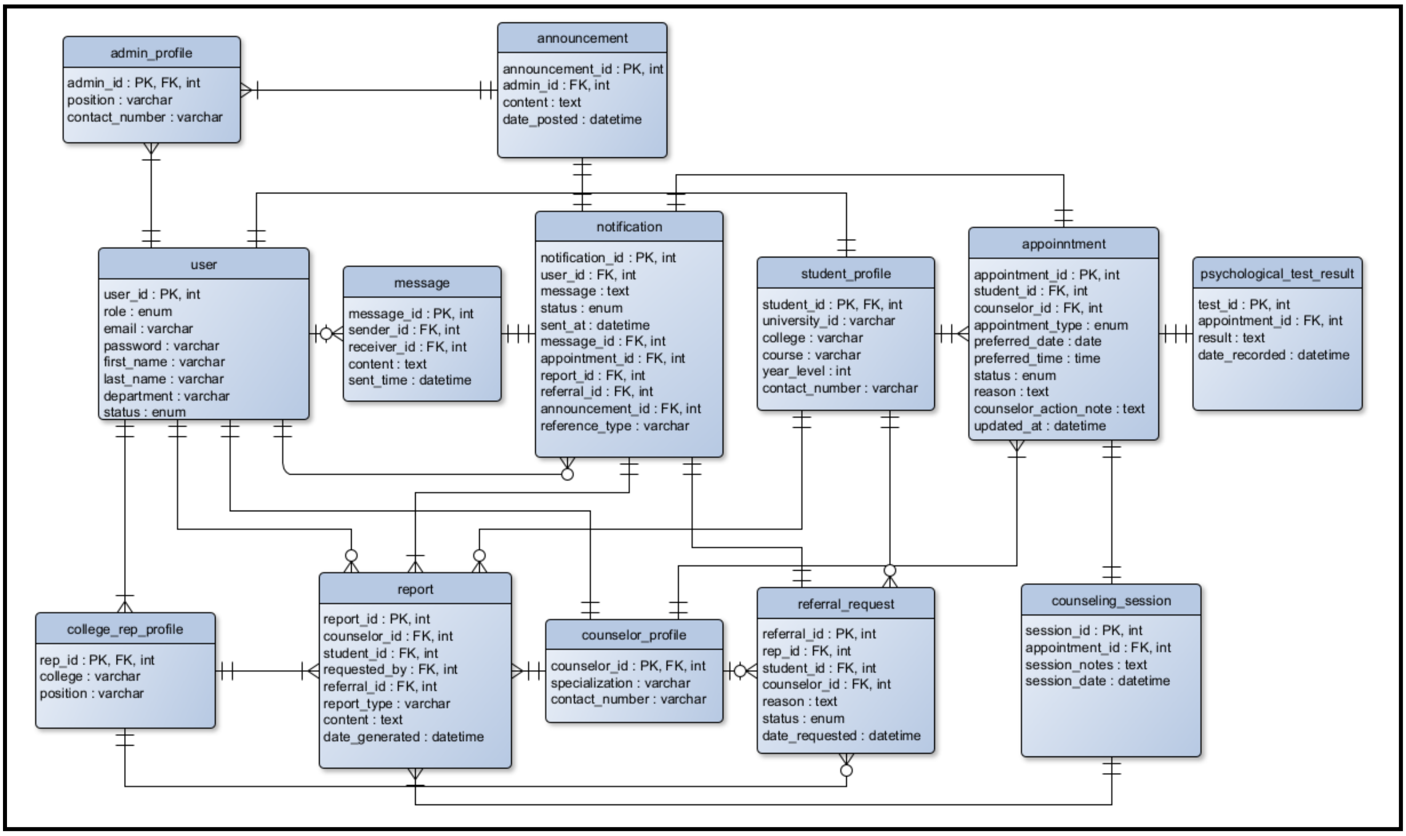

Figure 3.2.4 Entity-Relationship Diagram

3.2.5 Architectural Design

Figure 3.2.5 illustrates the system architecture of the CounseLink web-based guidance counseling system. The system will follow a three-tier architecture composed of the Client Layer, Application Layer, and Data Layer. The Client Layer consists of the student, guidance counselor, administrator, and college representative who interact with the system through secure web access. The Application Layer handles all system processes such as authentication, appointment scheduling, counseling record management, psychological test management, reporting, and notification services. The Data Layer stores all user data, appointment records, counseling information, test results, and system logs in a centralized and secure database. This architecture ensures secure, efficient, and reliable processing of counseling services within the university.

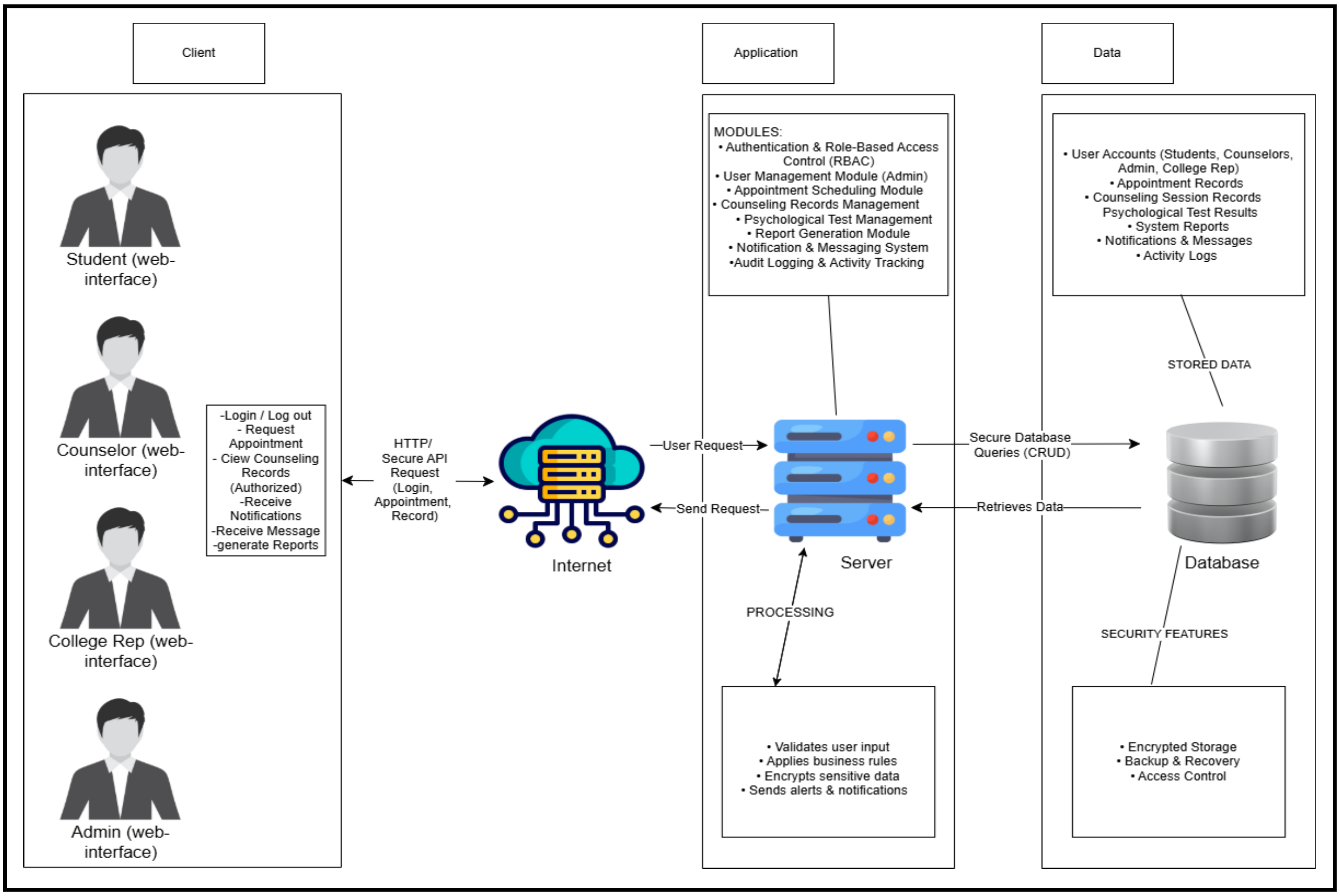

Figure 3.2.5 Architectural Design

3.3 Development

Developing of the proposed system specified hardware and software resources were required. The developer's hardware and software specification below.

3.3.1 Software Specification

The software development tools for this study such as the software component, minimum Requirements and recommended are presented in Table 3.3.1.1 to Table 3.3.1.5.

Table 3.3.1.1 Developer's software requirements

Software Component	Minimum Requirements	Recommended
Operating system	Window 10/11	Windows 11 / Ubuntu 20.04 LTS
DBMS	Rational (MySQL, Oracle)	MySQL 8.0 / MariaDB
Programming Language	HTML, CSS, JavaScript, React.js (Frontend), PHP/Nodes.js (Backend)	Same with latest stable versions
Programming Environment	Visual Studio Code	Visual Studio Code
Web server	XAMPP	XAMPP / Nginx
Browser	Chrome, Edge, Firefox for testing and debugging	Google Chrome (Latest)

Framework/Library	React, Bootstrap or Tailwind	React + Tailwind CSS

Table 3.3.1.2 Student's software requirements

Software Component	Minimum Requirements	Recommended
Operating system	Android version 10-12	Android 13+
Web Browser	Chrome/ Firefox/ Edge	Google Chrome (Latest)
Internet Connection	Required for online access	Stable 4G/5G / WiFi
Device	Smartphone or tablet	Smartphone
Screen Resolution	Minimum 720p	Minimum 1080p
Email Client	Gmail	Gmail (Latest App)

Table 3.3.1.3 Admin's software requirements

Software Component	Minimum Requirements	Recommended
Operating system	Window 7 or 10	Window 11
Web Browser	Chrome / Firefox / Edge	Google Chrome (Latest)
Internet Connection	Required for online access	Wired or Fiber Internet
Device	Desktop or Laptop	Desktop
Screen Resolution	Minimum 1080p	1920 x 1080 Full HD

Table 3.3.1.4 Counselor's software requirements

Software Component	Minimum Requirements	Recommended
Operating system	Window 10/11	Windows 11
Web Browser	Chrome / Firefox / Edge	Google Chrome (Latest)
Internet Connection	Required for online access	Fiber / Broadband Internet
Device	Desktop or Laptop	Laptop
Screen Resolution	Minimum 1080p	1920 × 1080 Full HD
Email Client	Gmail	Gmail (Latest App)

Table 3.3.1.5 College representative's software requirements

Software Component	Minimum Requirements	Recommended
Operating system	Window 10/11	Windows 11
Web Browser	Chrome / Firefox / Edge	Google Chrome (Latest)
Internet Connection	Required for online access	Fiber / Broadband Internet
Device	Desktop or Laptop	Desktop
Screen Resolution	Minimum 1080p	1920 × 1080 Full HD
Email Client	Gmail	Gmail (Latest App)

3.3.2 Hardware Specification

The researcher presents the minimum hardware resources necessary for CounseLink system and presented in Table 3.3.2.1 to Table 3.3.2.5.

Table 3.3.2.1 Developer's hardware requirements

Software Component	Minimum Requirements	Recommended
Processor	Intel core i5/i7	Intel Core i7 / Ryzen 7
Memory (RAM)	8-16 GB RAM	16-32 GB
Storage	128 GB SSD	1 TB SSD
Display	1920 x 1080 resolution	4k Display
Other peripherals	Keyboard, mouse, external monitor	UPS, Mechanical Keyboard

Table 3.3.2.2 Admin's hardware requirements

Software Component	Minimum Requirements	Recommended
Processor	Intel core i5/i7	Intel Core i7
Memory (RAM)	8-16 GB	16 GB
Storage	128GB HDD/SSD	512 GB SSD
Display	1920 x 1080 resolution	4K Display
Other peripherals	Keyboard and mouse	Printer, UPS

Table 3.3.2.3 Student's hardware requirements

Software Component	Minimum Requirements	Recommended
Processor	2.0 GHz Snapdragon 675 Octa-core	Snapdragon 778G
Memory (RAM)	8 GB	12 GB
Storage	64-128 GB	256 GB
Display	LCD or OLED	Full HD + OLED
Other peripherals	None	None

Table 3.3.2.4 Counselor's hardware requirements

Software Component	Minimum Requirements	Recommended
Processor	Intel Core i3 / i5 (or equivalent laptop)	Intel Core i5/i7
Memory (RAM)	4-8 GB	16 GB
Storage	128 GB SSD or HDD	512 GB SSD
Display	1920 x 1080 resolution	2K / 4K Display
Other peripherals	Keyboard, Mouse	Printer, Webcam

Table 3.3.2.5 College representative's hardware requirements

Software Component	Minimum Requirements	Recommended
Processor	Intel Core i3 / i5	Intel Core i7
Memory (RAM)	4-8 GB	16 GB
Storage	128 GB SSD or HDD	512 GB SSD
Display	1920 x 1080 resolution	2K / 4K Display

Other peripherals	Keyboard, Mouse	Printer, UPS

3.4 Testing

This testing plan will ensure that the system functions correctly, meets user requirements, and provides an acceptable level of usability and performance. This section discusses the respondents, procedures, and tools used during the testing phase.

3.4.1 Respondents and Locale

A total of 40 respondents will be participating in the system testing of CounseLink. These respondents will compose of actual end-users of the system and will be grouped as follows:

- 30 Students who will used the system to request guidance counseling appointments, view psychological test results, receive notifications, and monitor their counseling history.
- 5 Guidance Counselors who will manage appointments, document counseling sessions, upload psychological test results, and generate counseling reports.
- 5 College Representatives who will request student counseling data, monitor approved reports, and submit referrals for students.

The locale of the study will be the Mindanao State University – Marawi City, particularly the Division of Student Affairs, classrooms, dormitory, and faculty offices where end-users commonly interact with the system.

3.4.2 Procedure

The System Usability Scale and functional testing will follow this procedure:

1. Orientation - Respondents will be briefed on the purpose of the testing and will be given a short demonstration of the system.
2. Hands-on System - The researcher will use each respondent to perform actual tasks based on their user role, such as:
    - Students: requesting appointments, updating profiles, etc.
    - Counselors: approving appointments, managing student records, etc.
    - College reps: requesting counseling data, counseling referral, etc.
3. Completion of Test Cases Specific scenarios will be executed to verify correct system behavior.
4. Answering the Questionnaire - Respondents need to answer a User Acceptance Evaluation Form using a SUS to assess usability, functionality, accuracy, interface design, and overall satisfaction.
5. Feedback Collection - Additional comments and suggestions will be gathered for final system refinement.
6. Analysis and Documentation - Results will be processed to evaluate if the system met the required performance and acceptance level.

3.4.3 Test Cases

Table 3.4.3.1: Test Cases for Student

Test Case	Precondition	Description	Test Step	Expected Result	Actual Result	Status (Pass/Fail)	Postcondition
TC-1	Student is on registration page	Student registers an account	Open signup → Enter details → Submit	Account successfully created		(Pass/Fail)	Student account is available
TC-2	Student account already exists	Student Logs in	Enter valid credentials → Login	Student redirected to dashboard		(Pass/Fail)	Student session started
TC-3	Student is logged in	Student updates profile	Edit profile → Save	Profile updated successfully		(Pass/Fail)	Profile changes saved
TC-4	Student is logged in	Student requests counseling appointment	Select date & counselor → Submit	Request saved as Pending		(Pass/Fail)	Appointment request recorded
TC-5	Student has existing appointment	Student views appointment status	Open appointment list	Status displayed correctly		(Pass/Fail)	Appointment status viewed
TC-6	Student has pending/approved request	Student reschedules appointment	Edit request → Submit	New schedule saved		(Pass/Fail)	Updated schedule stored
TC-7	Student is logged in	Student requests psychological test	Click Request Test → Submit	Request successfully sent		(Pass/Fail)	Test request recorded
TC-8	Result is uploaded by counselor	Student views psychological test result	Open test results	Result file opens/downloads		(Pass/Fail)	Result accessed
TC-9	Student has past counseling record	Student views counseling history	Open history section	Records displayed correctly		(Pass/Fail)	History reviewed
TC-10	Student is logged in	Student sends message to counselor	Compose → Send	Message delivered successfully		(Pass/Fail)	Message stored
TC-11	Student has appointment update	Student receives system notification	Wait for update	Notification appears		(Pass/Fail)	Notification displayed

TC-12	Student is logged in	Student logs out	Click logout	Session terminated securely		(Pass/Fail)	User session ended

Table 3.4.3.2: Test Cases for Counselor

TC-13	Counselor account exists	Counselor logs in	Enter credentials → Login	Dashboard loads successfully		(Pass/Fail)	Counselor session started
TC-14	There are appointment requests	Counselor views appointment requests	Open request list	Requests displayed		(Pass/Fail)	Requests reviewed
TC-15	Pending request exists	Counselor approves appointment	Select request → Approve	Status becomes Approved and student notified		(Pass/Fail)	Appointment confirmed
TC-16	Pending request exists	Counselor rejects appointment	Select request → Reject	Status becomes Rejected		(Pass/Fail)	Request declined
TC-17	Appointment already approved	Counselor reschedules appointment	Select request → Reschedule	Student notified of new schedule		(Pass/Fail)	Updated schedule stored
TC-18	Session is completed	Counselor adds counseling record	Enter session notes → Save	Record saved successfully		(Pass/Fail)	Session recorded
TC-19	Existing record is available	Counselor edits counseling record	Update record → Save	Record updated		(Pass/Fail)	Record updated
TC-20	Existing record is selected	Counselor deletes counseling record	Delete → Confirm	Record deleted		(Pass/Fail)	Record removed
TC-21	Student has requested test	Counselor uploads test result	Upload file → Submit	File uploaded & student notified		(Pass/Fail)	Result stored
TC-22	Counseling record exists	Counselor generates counseling report	Click Generate	Report generated		(Pass/Fail)	Report available
TC-23	Generated report exists	Counselor sends report to college rep	Select report → Send	College Rep receives report		(Pass/Fail)	Report shared
TC-24	Student record exists	Counselor views	Select student	Records displayed		(Pass/Fail)	History accessed

		student counseling history					
TC--25	College rep has sent request	Counselor receives college request	Open request panel	Request displayed		(Pass/Fail)	Request acknowledged
TC-26	Student has sent a message	Counselor replies to student message	Type reply → Send	Message delivered		(Pass/Fail)	Reply stored
TC-27	Counselor is logged in	Counselor logs out	Click logout	Session terminated securely		(Pass/Fail)	Counselor session ended

Table 3.4.3.3: Test Cases for Administrator

TC-28	Admin account exists	Admin logs in	Enter credentials → Login	Dashboard displayed		(Pass/Fail)	Admin session started
TC-29	Admin is logged in	Admin creates student account	Enter details → Save	Account created		(Pass/Fail)	Student added
TC-30	Admin is logged in	Admin creates counselor account	Enter details → Save	Account created		(Pass/Fail)	Counselor added
TC-31	Admin is logged in	Admin creates college rep account	Enter details → Save	Account created		(Pass/Fail)	College rep added
TC-32	User account exists	Admin edits user account	Edit → Save	Account updated		(Pass/Fail)	User updated
TC-33	Active user account exists	Admin deactivates user account	Select → Deactivate	Account disabled		(Pass/Fail)	User access revoked
TC-34	Counselor and college exist	Admin assigns counselor to college	Select counselor & college	Assignment saved		(Pass/Fail)	Assignment completed
TC-35	System has log records	Admin views audit logs	Open logs	Logs displayed		(Pass/Fail)	Audit reviewed
TC-36	Admin is logged in	Admin posts announcement	Type → Post	Announcement visible		(Pass/Fail)	Announcement published

TC-37	System data exists	Admin generates system report	Click Generate	Report generated		(Pass/Fail)	System report available
TC-38	System is running	Admin backs up database	Click Backup	Backup created		(Pass/Fail)	Backup stored
TC-39	Unauthorized user attempts access	System enforces RBAC	Student attempts admin page	Access denied		(Pass/Fail)	Unauthorized access blocked
TC-40	Admin is logged in	Admin logs out	Click logout	Session ended		(Pass/Fail)	Admin session ended

Table 3.4.3.4: Test Cases for College Representative

TC-41	College rep account exists	College rep logs in	Enter credentials → Login	Dashboard loads		(Pass/Fail)	College rep session started
TC-42	Student has counseling record	College rep requests counseling report	Select student → Request	Request sent to counselor		(Pass/Fail)	Request recorded
TC-43	Approved report exists	College rep views approved reports	Open reports section	Reports displayed		(Pass/Fail)	Reports reviewed
TC-44	Approved report exists	College rep downloads report	Click download	File downloaded		(Pass/Fail)	File saved locally
TC-45	Student exists in system	College rep submits student referral	Enter details → Submit	Referral sent		(Pass/Fail)	Referral recorded
TC-46	Referral was previously sent	College rep views referral status	Open referral list	Status displayed		(Pass/Fail)	Status checked
TC-47	Report is available	College rep prints counseling report counseling report	Click Logout	Report printed		(Pass/Fail)	Printed copy available
TC-48	College rep is logged in	College rep logs out	Click Logout	Session terminated		(Pass/Fail)	College rep session ended

3.4.4 Testing Tools

The database application system will be tested and evaluated using SUS (System Usability Scale), it is a reliable tool for measuring usability. It consists of 10 item questionnaires with five response options for respondents; from strongly agree to strongly disagree. Originally created by John Brooke in 1989, it will allow us to evaluate this capstone project.

The Categorization that provided (1 to strongly agree, 2 to agree, 3 to neutral, 4 to disagree, and 5 to strongly disagree) into the traditional 5-point Likert scale will use in the System Usability Scale (SUS).

Strongly Disagree (5 in your scale)
Disagree (4 in your scale)
Neutral (3 in your scale)
Agree (2 in your scale)
Strongly Agree (1 in your scale)

Now, with the traditional 5-point Likert scale, we will use the categorizations mentioned earlier.

Excellent (SUS Score >= 80):

Users who answered, "Strongly Agree" (1) to most of the questions. They find the system highly usable.

Good (SUS Score 68 to 79):

Users who answered "Agree" (2) to most of the questions. They find the system generally usable with some room for improvement.

Average (SUS Score 51 to 67):

Users who answered "Neutral" (3) to most of the questions. They are somewhat undecided or have mixed feelings about the system's usability.

Poor (SUS Score <= 50):

Users who answered "Disagree" (4) or "Strongly Disagree" (5) to most of the questions. They find the system difficult to use, and there are significant usability issues.

When SUS is used, participants are asked to score the following 10 items with one of five responses that range from strongly agree to strongly disagree:

1. I think that I would like to use this system frequently.
2. I found the system unnecessarily complex.
3. I thought the system was easy to use.
4. I think that I would need the support of a technical person to be able to use this system.
5. I found the various functions in this system were well integrated.
6. I thought there was too much inconsistency in this system.
7. I would imagine that most people would learn to use this system very quickly.
8. I found the system very cumbersome to use.
9. I felt very confident using the system.
10. I need to learn a lot of things before I could get going with this system.

The calculation process of the SUS score starts by summing the score contribution from each item. Each item's score contribution ranges from 0 to 4. For items 1, 3, 7, and 9 (the positive worded items), the score contribution is the scale position minus 1. For items 2, 4, 6, 8 and 10 (the negatively worded items), the contribution score is 5 minus the scale position. To obtain the overall value of SUS, the sum of the score of all items is taken and then multiplied to 2.5 the SUS scores ranged between 0 to 100 (Brooke, 1986).

3.4.4.1 Alpha Testing

This section, the database application system will be tested some programmers, professors or seniors in the College of Information Computing Sciences to verify the execution of the system and there was no error occurred during run on this system and run smoothly.

3.4.4.2 Beta Testing

To test efficiency and accuracy. These are parts that will be tested in our client, which is the Head unit of the Guidance Counseling Section of the Division of Student Affairs. They will determine the efficiency and the extent of the acquired benefit the system can offer compared to manually doing the institution's transaction.

3.4.4.3 Evaluation Method

3.4.4.4 System Usability Scale (SUS) Test Questionnaire

The system Usability Scale (SUS) Test Questionnaire will be used by the researcher to evaluate the testing period of the system. SUS is one of the most efficient ways of gathering statistically valid data and giving your system a clear and reasonably precise score. The system Usability scale is a Likert Scale Which Includes 10 Questionnaires which users of the system will answer.

3.4.4.5 Likert Scale

Likert Scale rating system, will be used in questionnaires, is designed to measure people's attitudes, opinions, or perceptions. Subjects choose from a range of possible responses to a specific question or statement; response typically include "Strong agree", "Agree", "Neutral", "Disagree" and "Strong Disagree". Often, the categories of response are coded numerically, in which case the numerical values must defined for that specific study such as one (1) = Strong Agree, two (2) = Agree, Three (3) = Neutral, Four (4) = Disagree, Five (5) = Strong Disagree, (Jamieson, 2019). The researcher chose the 5- point Likert scale for the evaluation of weighted mean of each result.

3.4.4.6 Weighted Mean Formula

The researcher will use the weighted mean formula to validate the total of each questionnaire scoring from respondent on System Usability Scale (SUS) assessment tool and the formula also used for the result SUS to have an

overall result for the validation of the acceptability of the system. The following formula for getting the weighted means is as follows:
W: weighted mean
w_i: weighted applied to x values
n: number of terms to be averaged
X_i: data values to be averaged

$$\bar{x} = \frac{\sum_{i=1}^{n} (x_i * w_i)}{\sum_{i=1}^{n} w_i}$$

Figure 3.4.4.6 Weighted means Formula

3.4.4.6.1 Weighted Mean on 5-point Likert Scale

To get the total weighted mean of each average from the System Usability Scale (SUS) Assessment tools and for the overall result from SUS Assessment tool and SUS questionnaires the researcher scaled it based on their total weighted mean. Below is the general guideline on weighted Mean Average for 5 Likert Scalability interpretation:

Table 3.4.4.6.1.1 Scalability of weighted Mean on 5 Likert Scale

LEGEND	ADJECTIVE	SUS SCORE
1	Strongly Agree	1.00 – 1.80
2	Agree	1.81 – 2.60
3	Neutral	2.61 – 3.40
4	Disagree	3.41 – 4.20
5	Strong Disagree	4.21 – 5.00

For the reversed item in Likert Scale, below was the given Formula for reversed scoring:

Table 3.4.4.6.1.2 Reversed item Scoring on 5 Likert Scale

REVERSED SCORING	LIKERT SCALE
5	If (Strongly Agree = 1)

4	If (Agree = 2)
3	If (Neutral = 3)
2	If (Disagree = 4)
1	If (Strong Disagree = 5)

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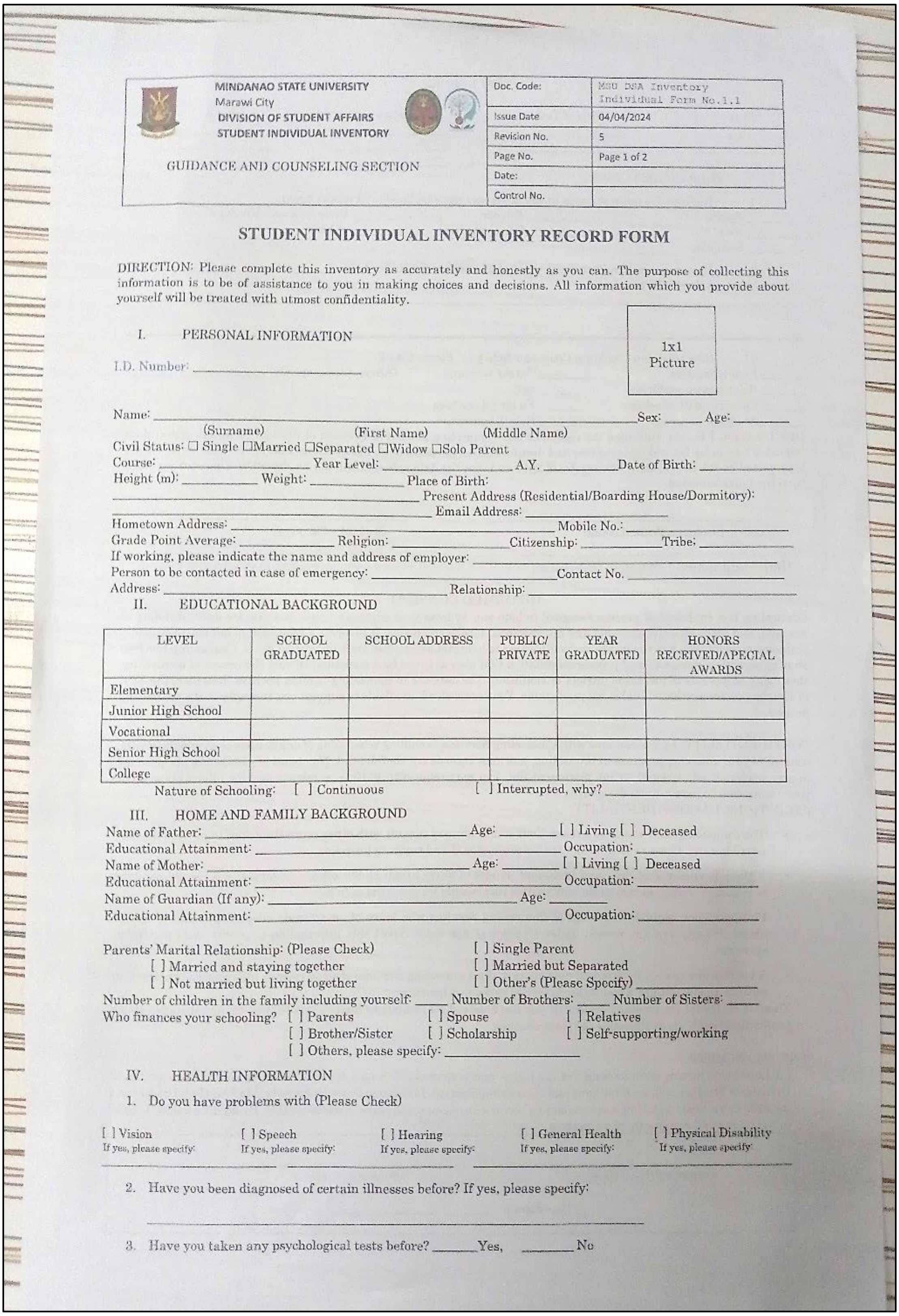
APPENDICES

Appendix A: Glossary

- **Counseling** - A professional process where trained guidance counselors help students manage personal, academic, or emotional concerns.
- **Guidance and Counseling Services**- University support services that provide emotional, psychological, academic, and social assistance to students.
- **Counseling Record** -A documented history of a student's counseling session, including concerns, interventions, recommendations, and follow-up actions.
- **Division of Student Affairs (DSA**) - The MSU–Marawi office responsible for managing student welfare, including guidance counseling, student policies, and psychosocial services.
- **Web-Based System** - A digital platform accessible through an internet browser that allows users to perform transactions and access services online.
- **Appointment Scheduling** - A system function that allows students to choose and reserve a counseling session time and counselor.
- **Psychological Testing** - Standardized tests administered by counselors to assess students' emotional, academic, or behavioral conditions.
- **Role-Based Access Control (RBAC**)- A security method that restricts system access based on a user's assigned role (student, counselor, admin, college representative).
- **Counseling Report** - A summary created by counselors that contains student concerns, progress, recommendations, and actions taken. It may also be shared with colleges for monitorin

Appendix B: Documents/ Sample Form



MINDANAO STATE UNIVERSITY Marawi City DIVISION OF STUDENT AFFAIRS STUDENT INDIVIDUAL INVENTORY GUIDANCE AND COUNSELING SECTION	Doc. Code:	MSU DSA Inventory Individual Form No.1.1
	Issue Date	04/04/2024
	Revision No.	5
	Page No.	Page 1 of 2
	Date:	
	Control No.	

STUDENT INDIVIDUAL INVENTORY RECORD FORM

DIRECTION: Please complete this inventory as accurately and honestly as you can. The purpose of collecting this information is to be of assistance to you in making choices and decisions. All information which you provide about yourself will be treated with utmost confidentiality.

I. PERSONAL INFORMATION

1x1 Picture

I.D. Number: ____________________

Name: ____ Sex: _____ Age: ________
(Surname) (First Name) (Middle Name)
Civil Status: ☐ Single ☐Married ☐Separated ☐Widow ☐Solo Parent
Course: ________________ Year Level: ______________ A.Y. __________ Date of Birth: ____________
Height (m): __________ Weight: __________ Place of Birth:
______________________________ Present Address (Residential/Boarding House/Dormitory):
______________________________ Email Address: ______________
Hometown Address: ______________________________ Mobile No.: ______________
Grade Point Average: ____________ Religion: ______________ Citizenship: __________ Tribe: __________
If working, please indicate the name and address of employer: ______________________________
Person to be contacted in case of emergency: __________________ Contact No. ______________
Address: ____________________________ Relationship: ____________________________

II. EDUCATIONAL BACKGROUND

LEVEL	SCHOOL GRADUATED	SCHOOL ADDRESS	PUBLIC/ PRIVATE	YEAR GRADUATED	HONORS RECEIVED/APECIAL AWARDS
Elementary					
Junior High School					
Vocational					
Senior High School					
College					

Nature of Schooling: [ ] Continuous [ ] Interrupted, why? ______________

III. HOME AND FAMILY BACKGROUND

Name of Father: ______________________________ Age: ________ [ ] Living [ ] Deceased
Educational Attainment: ______________________________ Occupation: ______________
Name of Mother: ______________________________ Age: ________ [ ] Living [ ] Deceased
Educational Attainment: ______________________________ Occupation: ______________
Name of Guardian (If any): ______________________________ Age: ________
Educational Attainment: ______________________________ Occupation: ______________

Parents' Marital Relationship: (Please Check) [ ] Single Parent
[ ] Married and staying together [ ] Married but Separated
[ ] Not married but living together [ ] Other's (Please Specify) ______________
Number of children in the family including yourself: ____ Number of Brothers: ____ Number of Sisters: ____
Who finances your schooling? [ ] Parents [ ] Spouse [ ] Relatives
[ ] Brother/Sister [ ] Scholarship [ ] Self-supporting/working
[ ] Others, please specify: ______________

IV. HEALTH INFORMATION

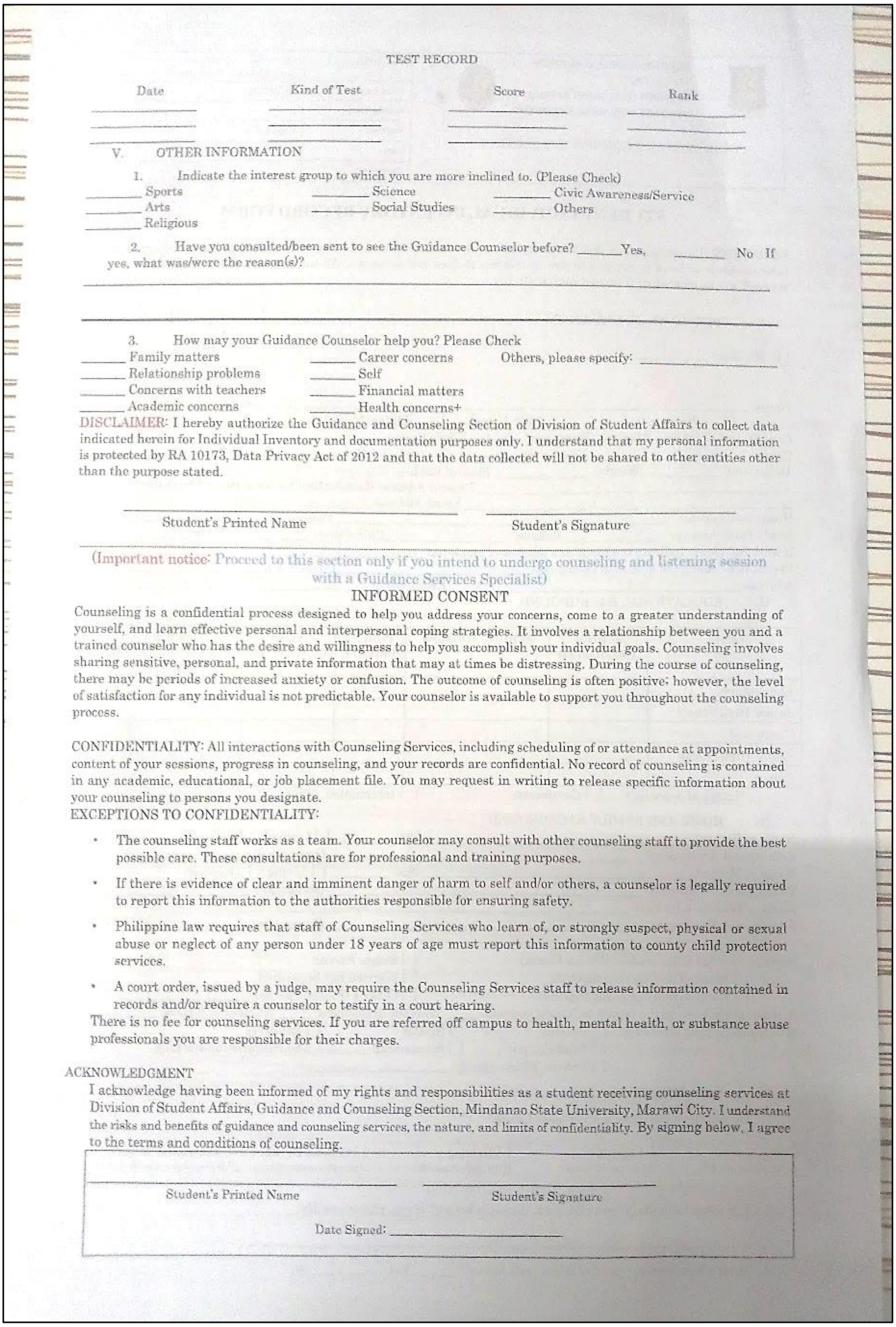
1. Do you have problems with (Please Check)

[ ] Vision If yes, please specify: ________ [ ] Speech If yes, please specify: ________ [ ] Hearing If yes, please specify: ________ [ ] General Health If yes, please specify: ________ [ ] Physical Disability If yes, please specify: ________

2. Have you been diagnosed of certain illnesses before? If yes, please specify:
____

3. Have you taken any psychological tests before? ______Yes, ______No

Form A: Student Individual Inventory Record Form



TEST RECORD

Date	Kind of Test	Score	Rank
________	________	________	________
________	________	________	________
________	________	________	________

V. OTHER INFORMATION

1. Indicate the interest group to which you are more inclined to. (Please Check)

_____ Sports _____ Science _____ Civic Awareness/Service
_____ Arts _____ Social Studies _____ Others
_____ Religious

2. Have you consulted/been sent to see the Guidance Counselor before? _____Yes, _____ No If yes, what was/were the reason(s)?

____

____

3. How may your Guidance Counselor help you? Please Check

_____ Family matters _____ Career concerns Others, please specify: __________
_____ Relationship problems _____ Self
_____ Concerns with teachers _____ Financial matters
_____ Academic concerns _____ Health concerns+

DISCLAIMER: I hereby authorize the Guidance and Counseling Section of Division of Student Affairs to collect data indicated herein for Individual Inventory and documentation purposes only. I understand that my personal information is protected by RA 10173, Data Privacy Act of 2012 and that the data collected will not be shared to other entities other than the purpose stated.

________________________ ________________________
Student's Printed Name Student's Signature

(Important notice: Proceed to this section only if you intend to undergo counseling and listening session with a Guidance Services Specialist)

INFORMED CONSENT

Counseling is a confidential process designed to help you address your concerns, come to a greater understanding of yourself, and learn effective personal and interpersonal coping strategies. It involves a relationship between you and a trained counselor who has the desire and willingness to help you accomplish your individual goals. Counseling involves sharing sensitive, personal, and private information that may at times be distressing. During the course of counseling, there may be periods of increased anxiety or confusion. The outcome of counseling is often positive; however, the level of satisfaction for any individual is not predictable. Your counselor is available to support you throughout the counseling process.

CONFIDENTIALITY: All interactions with Counseling Services, including scheduling of or attendance at appointments, content of your sessions, progress in counseling, and your records are confidential. No record of counseling is contained in any academic, educational, or job placement file. You may request in writing to release specific information about your counseling to persons you designate.

EXCEPTIONS TO CONFIDENTIALITY:

- The counseling staff works as a team. Your counselor may consult with other counseling staff to provide the best possible care. These consultations are for professional and training purposes.
- If there is evidence of clear and imminent danger of harm to self and/or others, a counselor is legally required to report this information to the authorities responsible for ensuring safety.
- Philippine law requires that staff of Counseling Services who learn of, or strongly suspect, physical or sexual abuse or neglect of any person under 18 years of age must report this information to county child protection services.
- A court order, issued by a judge, may require the Counseling Services staff to release information contained in records and/or require a counselor to testify in a court hearing.

There is no fee for counseling services. If you are referred off campus to health, mental health, or substance abuse professionals you are responsible for their charges.

ACKNOWLEDGMENT

I acknowledge having been informed of my rights and responsibilities as a student receiving counseling services at Division of Student Affairs, Guidance and Counseling Section, Mindanao State University, Marawi City. I understand the risks and benefits of guidance and counseling services, the nature, and limits of confidentiality. By signing below, I agree to the terms and conditions of counseling.

________________________ ________________________
Student's Printed Name Student's Signature

Date Signed: ________________

Form B: Consent Form (at the back of Form A)

Mindanao State University – Main Campus
Division of Student Affairs
Guidance and Counseling Section

MINDANAO STATE UNIVERSITY Marawi City DIVISION OF STUDENT AFFAIRS STUDENT INDIVIDUAL INVENTORY GUIDANCE AND COUNSELING SECTION	Doc. Code:	MSU DSA GCS Form 3.3
	Issue Date	04/04/2024
	Revision No.	5
	Page No.	Page 1 of 2
	Date:	
	Control No.	

STUDENT COUNSELING FORM

Student's Name:	Date:
Counselor's Name:	

1. REASON FOR COUNSELING:
   - ☐ Routine (☐1st ☐2nd ☐3rd ☐4th ☐5th ☐6th ☐7th ☐8th ... _____)
   - ☐ Student Initiated
   - ☐ Institute Initiated

   ☐ Identify reason: ______________________________
   ____
   ____

2. Goals:

3. Summary of Counseling/Key Points of Discussion:

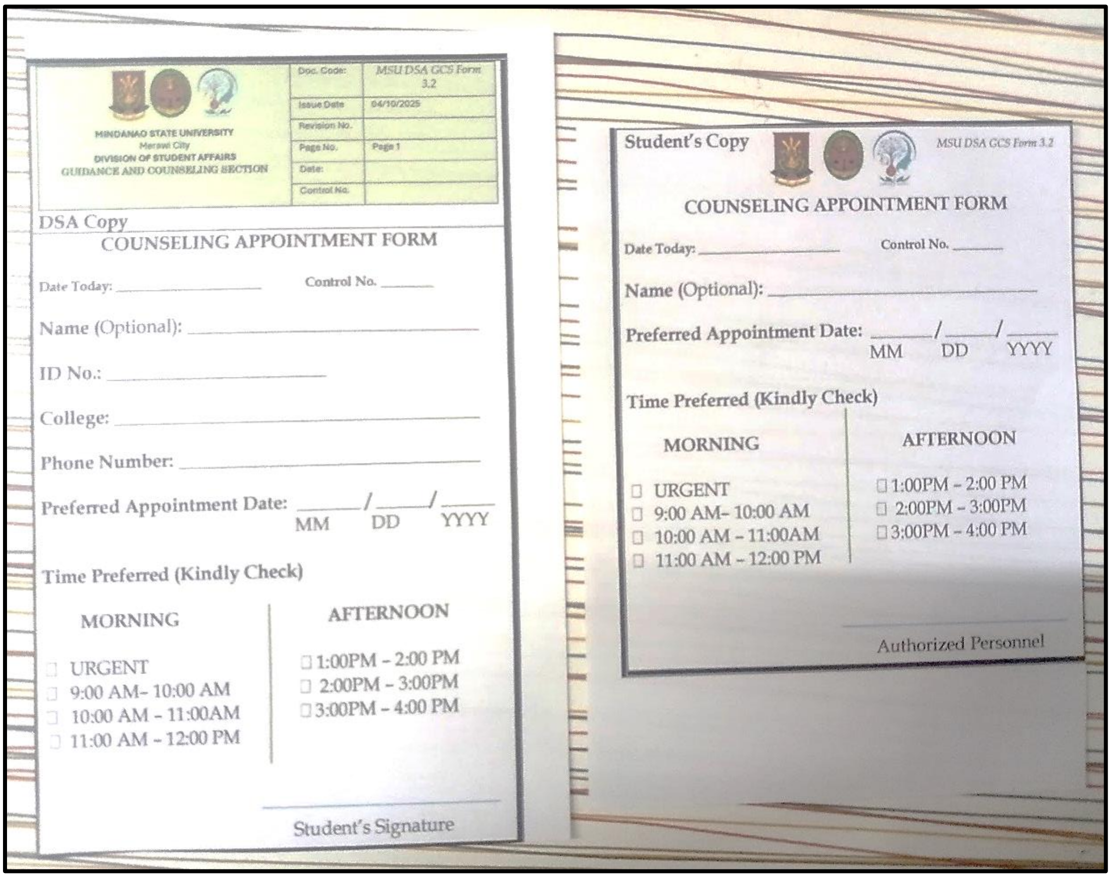
4. Plan of Action:

5. Counselor's comments:

6. Next counseling session (☐Follow-up ☐termination):

____
Signature of Counselor

Form C: Student counseling Form



MINDANAO STATE UNIVERSITY
Marawi City
DIVISION OF STUDENT AFFAIRS
GUIDANCE AND COUNSELING SECTION

Doc. Code:	MSU DSA GCS Form 3.2
Issue Date	04/10/2025
Revision No.	
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Date:	
Control No.	

DSA Copy

COUNSELING APPOINTMENT FORM

Date Today: ____________ Control No. ______

Name (Optional): ____________

ID No.: ____________

College: ____________

Phone Number: ____________

Preferred Appointment Date: ____/____/____
MM DD YYYY

Time Preferred (Kindly Check)

MORNING
- ☐ URGENT
- ☐ 9:00 AM- 10:00 AM
- ☐ 10:00 AM – 11:00AM
- ☐ 11:00 AM – 12:00 PM

AFTERNOON
- ☐ 1:00PM – 2:00 PM
- ☐ 2:00PM – 3:00PM
- ☐ 3:00PM – 4:00 PM

____
Student's Signature

Student's Copy MSU DSA GCS Form 3.2

COUNSELING APPOINTMENT FORM

Date Today: ____________ Control No. ______

Name (Optional): ____________

Preferred Appointment Date: ____/____/____
MM DD YYYY

Time Preferred (Kindly Check)

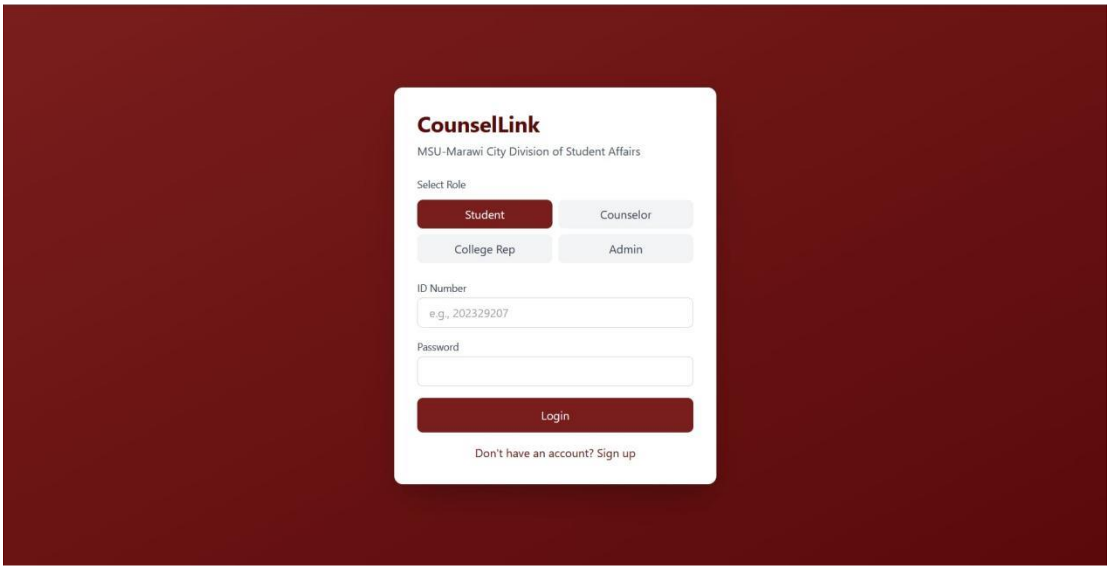
MORNING
- ☐ URGENT
- ☐ 9:00 AM- 10:00 AM
- ☐ 10:00 AM – 11:00AM
- ☐ 11:00 AM – 12:00 PM

AFTERNOON
- ☐ 1:00PM – 2:00 PM
- ☐ 2:00PM – 3:00PM
- ☐ 3:00PM – 4:00 PM

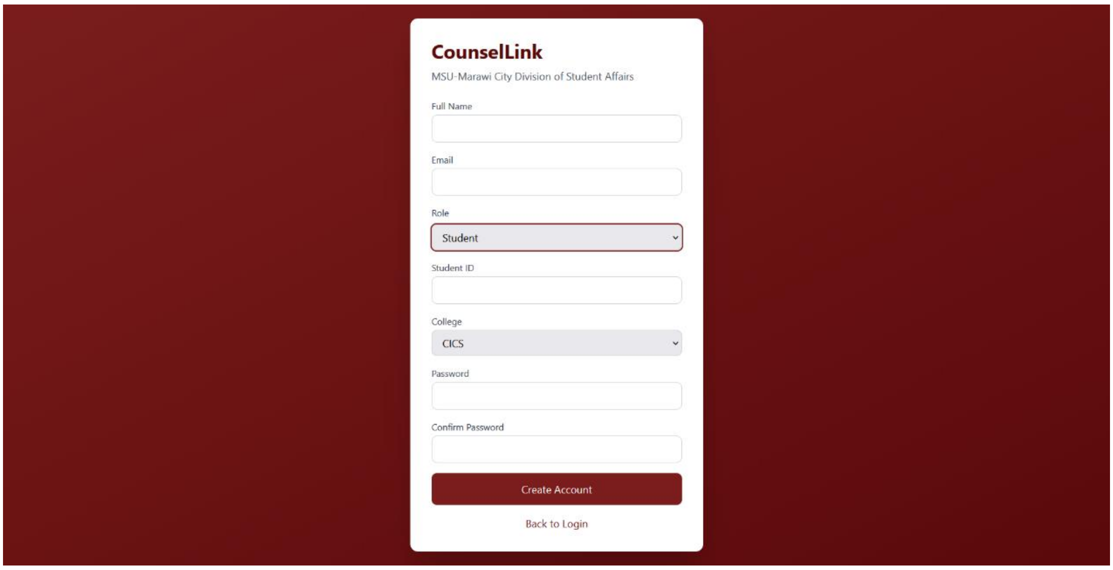
____
Authorized Personnel

Form D: Appointment Form (DSA copy and Student copy)

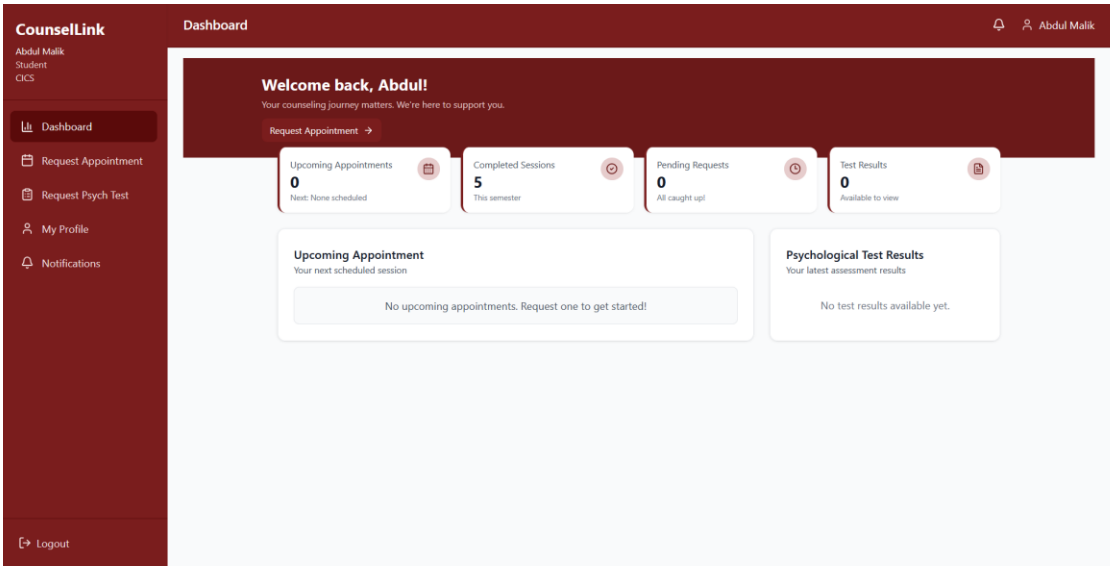
Appendix C: Prototype



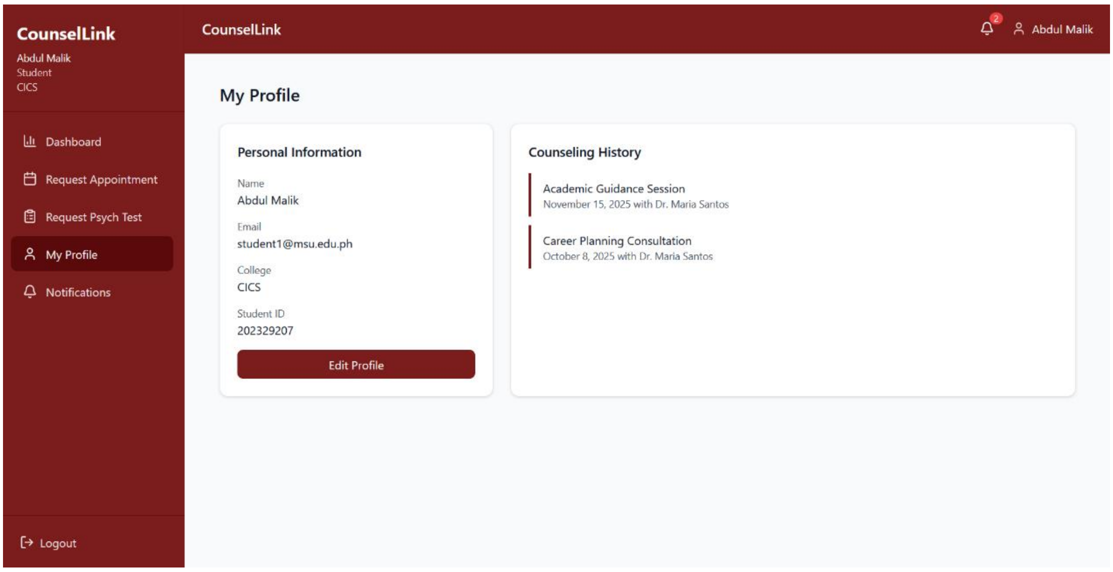
SS 1: Login Page Module



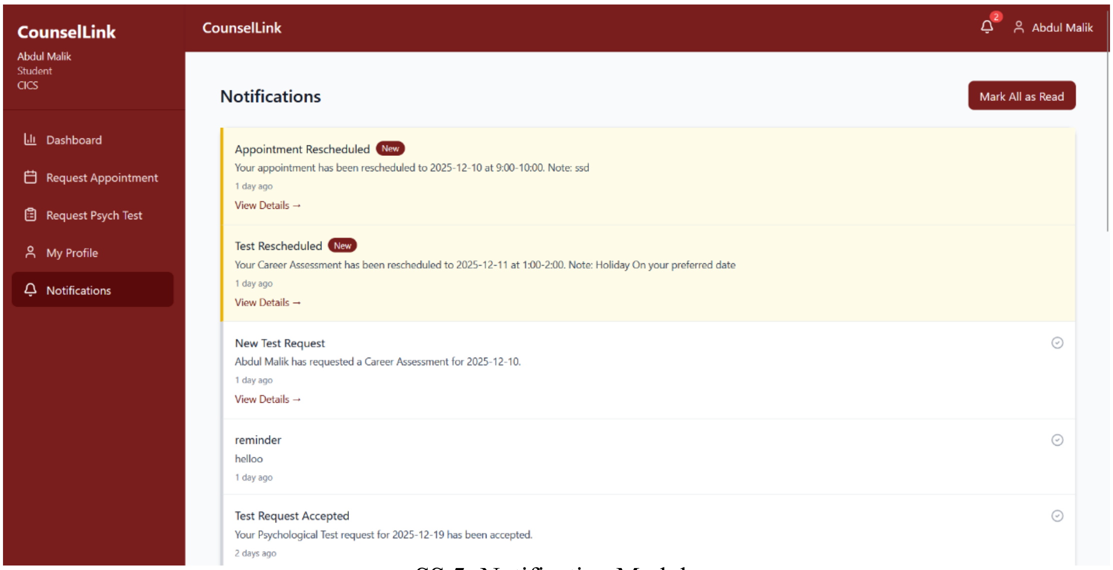
SS 2: Sign Up of Student



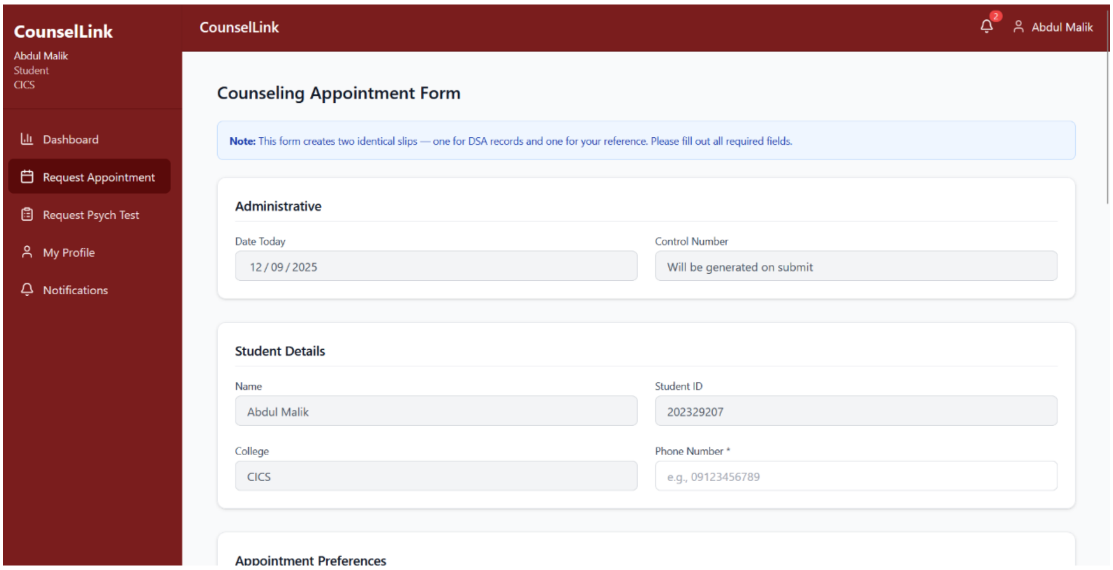
SS 3: Student Dashboard



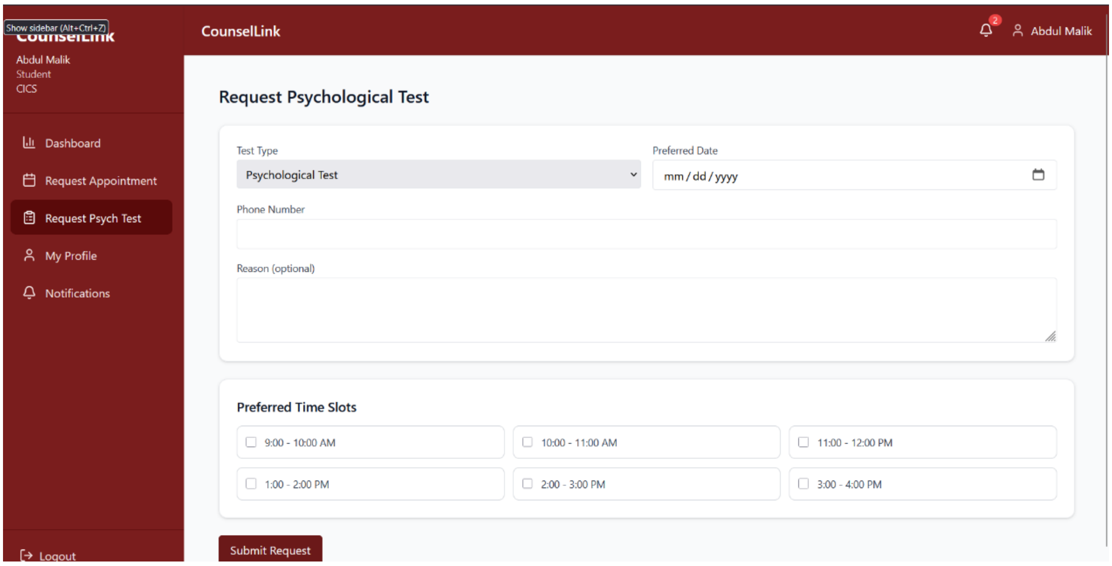
SS 4: Student Profile and Counseling History Module



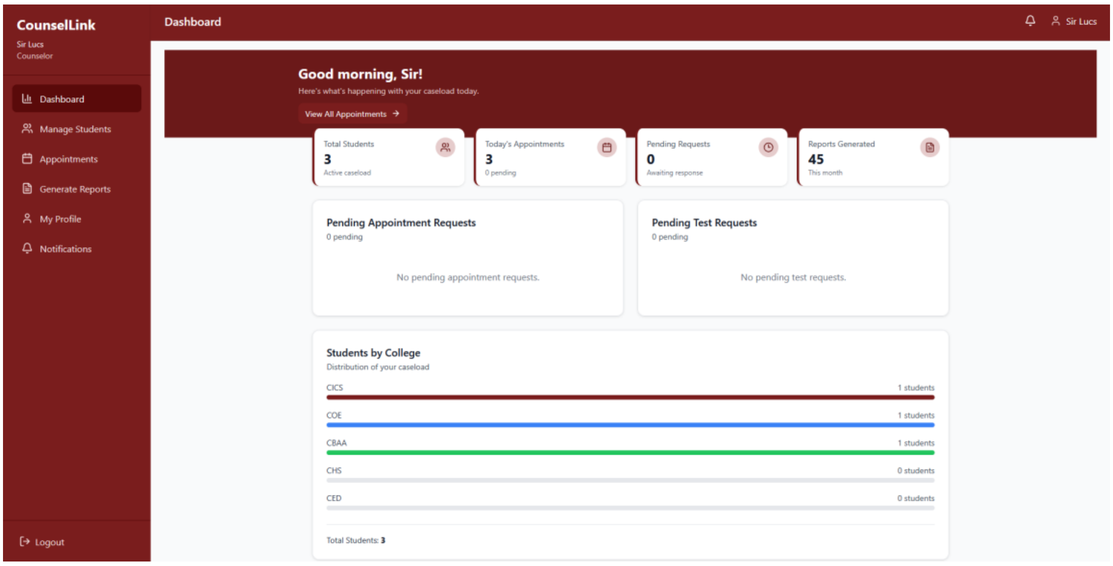
SS 5: Notification Module



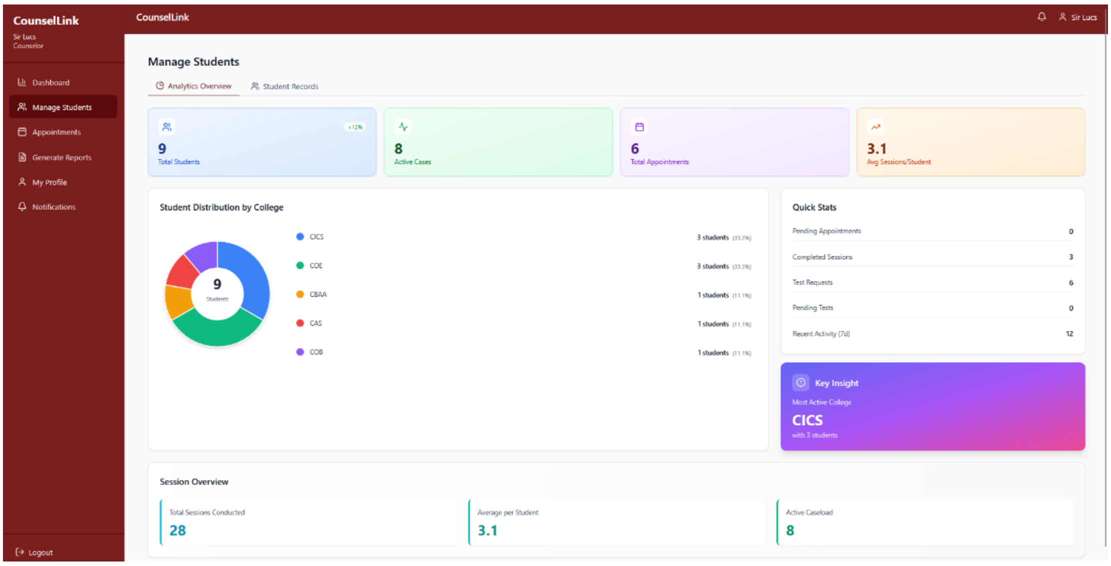
SS 6: Requesting Counseling Appointment



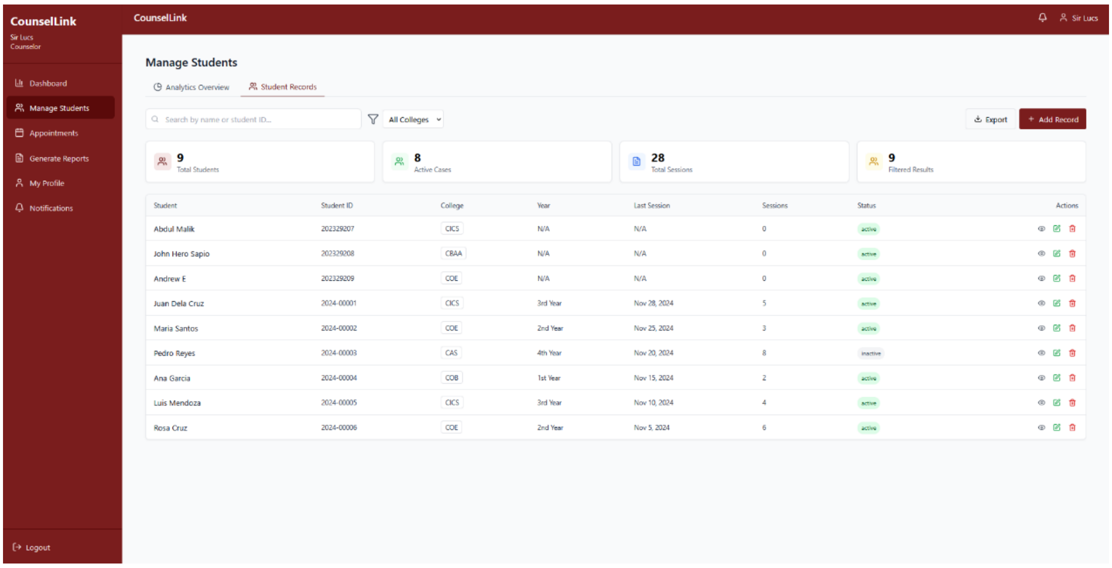
SS 7: Requesting Psychological Test Appointment



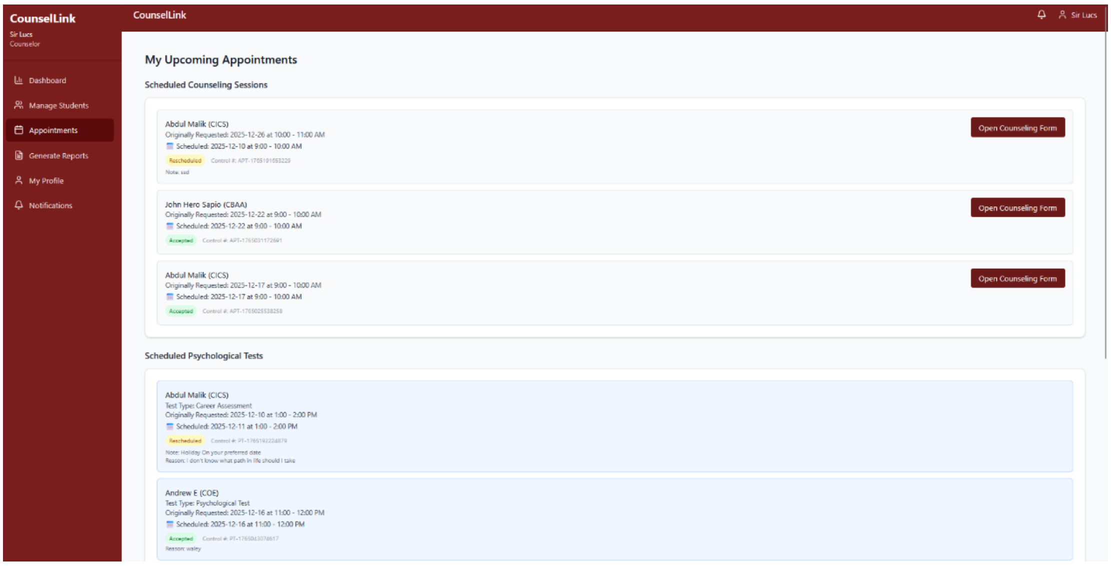
SS 8: Counselor Dashboard



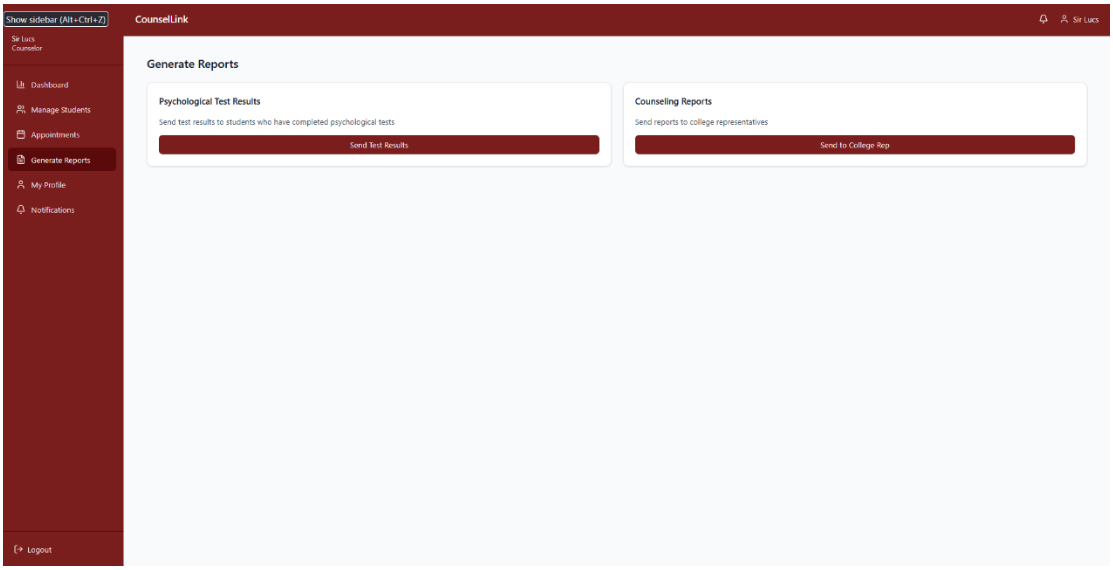
SS 9: Manage Student Module (Analytical Overview)



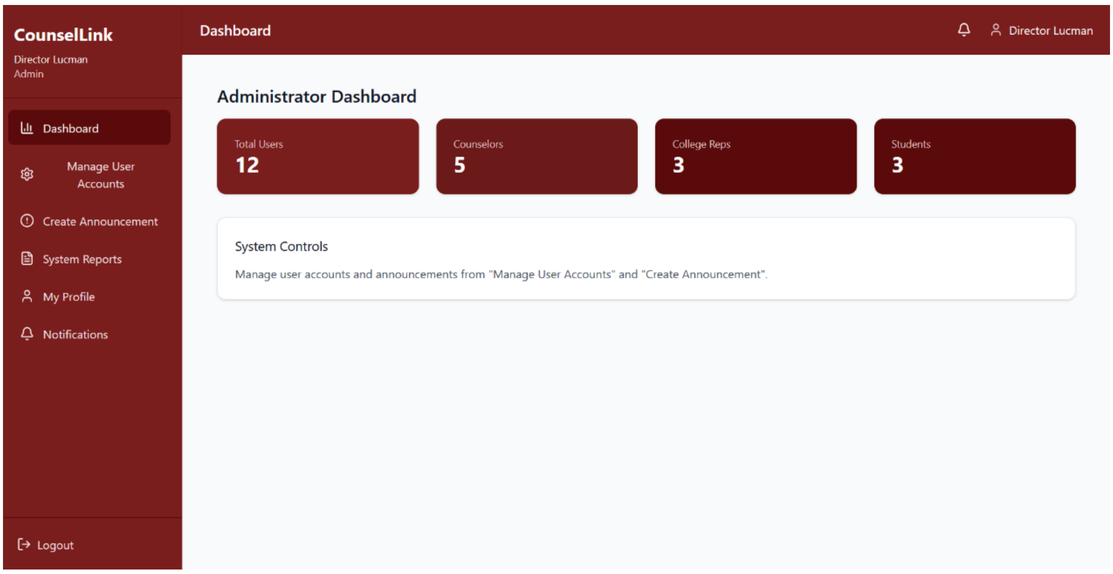
SS 10: Manage Student Module (Student Record)



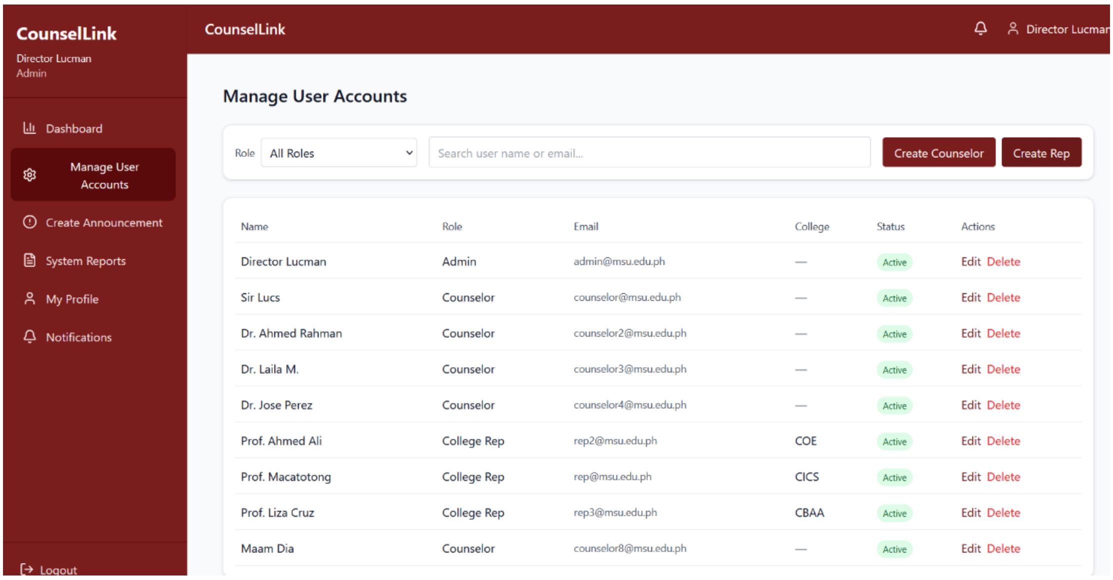
SS 11: Appointment Confirmation Module



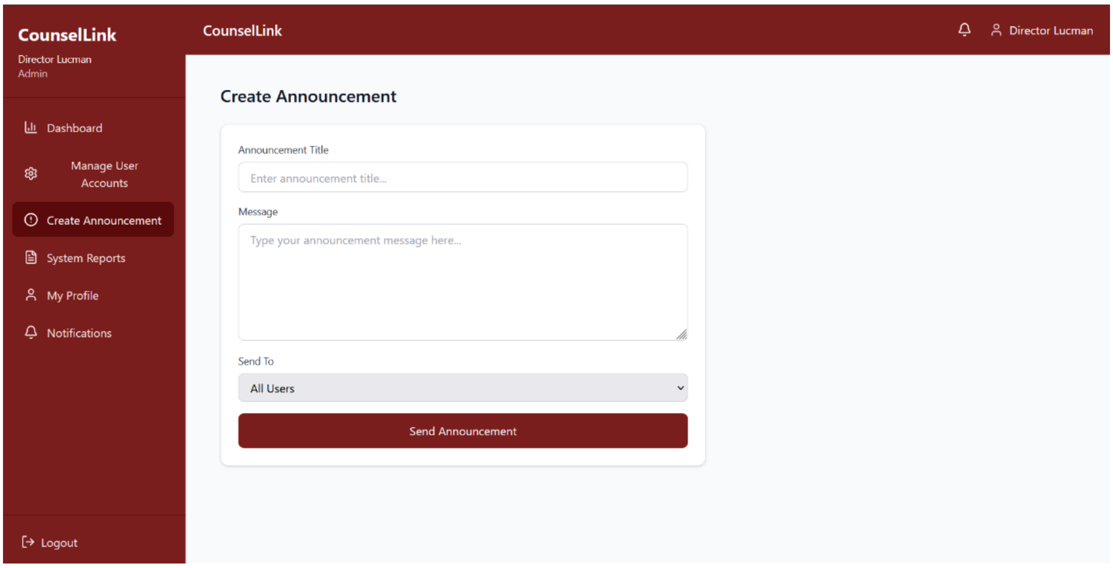
SS 12: Generate Report Module



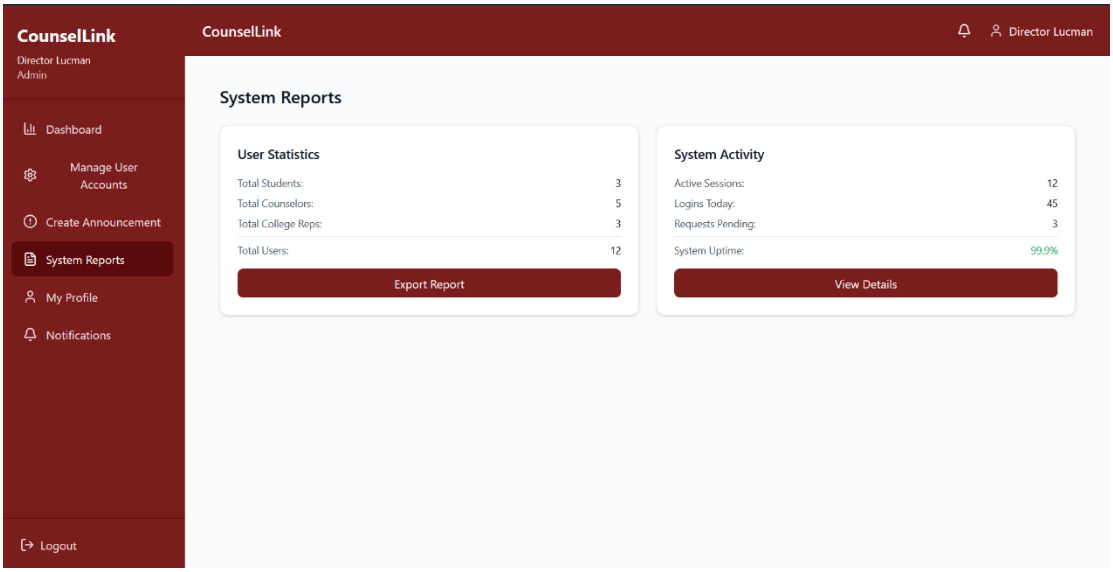
SS 13: Admin Dashboard



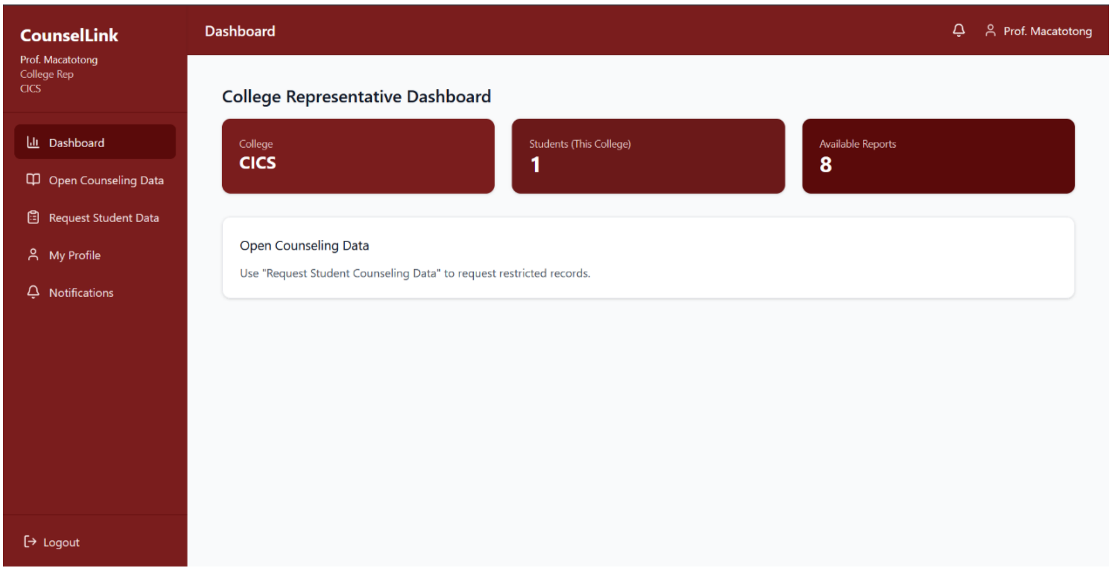
SS 14: Manage User Account Module



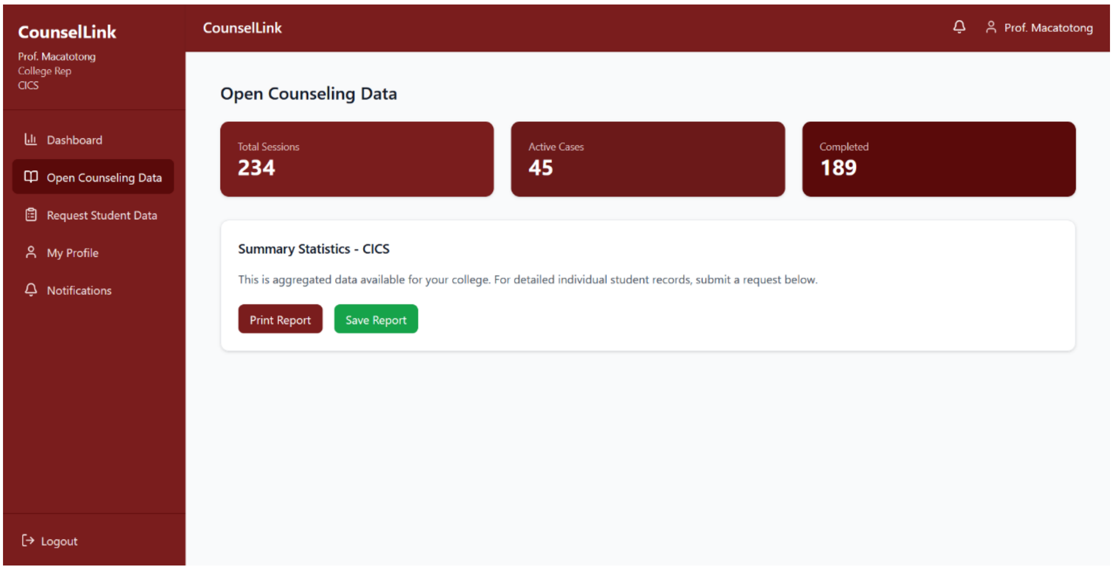
SS 15: Create Announcement Module



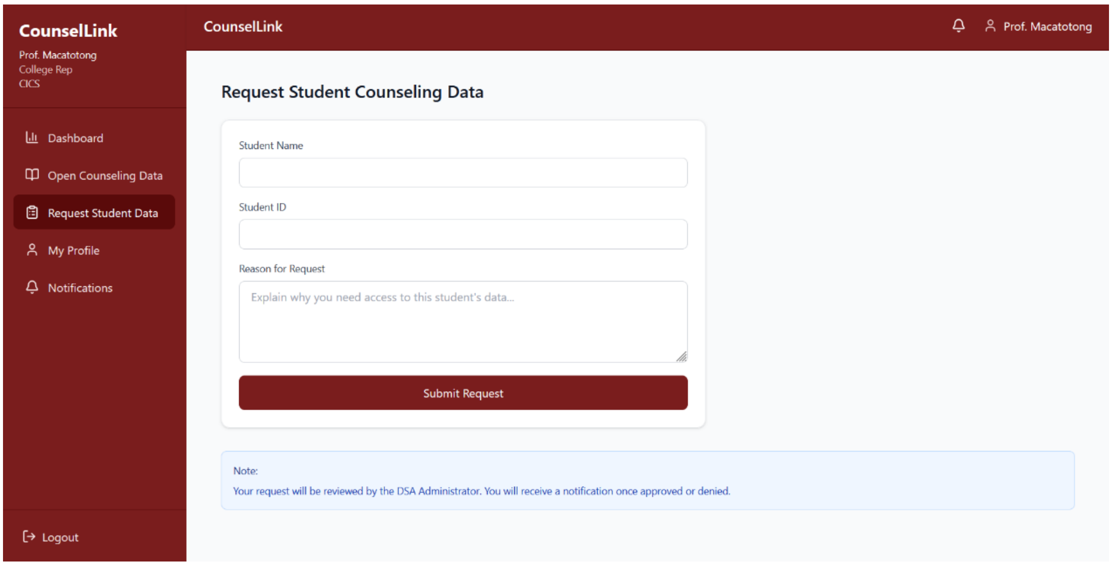
SS 16: System Reports



SS 17: College Representative Dashboard



SS 18: College Counseling Summary



SS 19: Request Student Counseling Data Report